

Model Analysis

Table of contents

1	Data analysis /Dataframe import and filters	2
1.1	DataFrame import and filters	2
1.2	Metadata	4
2	Model(s) analysis	5
2.1	Explanation legend tables	5
2.2	Repartition of train and test	6
2.2.1	External Split: X_train vs X_test	6
2.2.2	Internal CV Folds	6
2.3	Model : CatBoost	8
2.3.1	Gridsearch	8
2.3.2	Scatter	14
2.3.3	Scatter residuals	15
2.3.4	Learning curve	15
2.4	Model : LGBM	16
2.4.1	Gridsearch	16
2.4.2	Scatter	118
2.4.3	Scatter residuals	119
2.4.4	Learning curve	120
2.5	Sumup-table	121
3	Plot	121
3.1	Standalone plots	121
3.1.1	Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3	121
3.1.2	Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11	124
3.1.3	Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5	127
3.1.4	Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4	130
3.1.5	Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11	133
3.1.6	Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4	136
3.1.7	Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5	139

3.1.8	Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3	142
3.2	Compare plots	145
3.2.1	Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3	145
3.2.2	Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11	147
3.2.3	Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5	148
3.2.4	Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4	150
3.2.5	Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11	151
3.2.6	Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4	153
3.2.7	Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5	154
3.2.8	Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3	156

1 Data analysis /Dataframe import and filters

1.1 DataFrame import and filters

The data has been imported with the following(s) function(s):

```
def build_df():
    if flag_IR:
        df, features_engineered = import_data_IR()
        if flag_IR_maskP:
            df = mask_period_IR(df)
        if flag_IR_balR:
            df = mask_balanceHC_random_IR(df, n_samples=None)
    if flag_ER:
        df, features_engineered = import_data_ER()

    cols_to_check = target_col + features_X
    mask = np.isfinite(df[cols_to_check]).all(axis=1)
    df = df[mask]

    cols = list(dict.fromkeys(target_col + features_X +
                             feats_to_balance + feats_grouping +
                             feats_reference+ feats_postproc))

    df = df[cols]
    #df['campaign'] = "ALL"

    return df
```

```

def import_data_IR():
    if target_col[0] == "i_NH3":
        #file_path = "csv/innova/switched/df_signal_decomposition_NH3_current_db2.csv"
        file_path = "csv/innova/notswitched/df_signal_decomposition_current.csv"
    elif target_col[0] == "i_CO2":
        file_path = "csv/innova/switched/df_signal_decomposition_CO2_current.csv"
    else:
        raise ValueError("Target is not supported")

    df = pd.read_csv(file_path, sep=";")
    df["datetime"] = pd.to_datetime(df["datetime"])
    df.sort_values("datetime").reset_index(drop=True)

    if target_col[0] == "i_NH3":
        df["campaign"] = df["datetime"].apply(assign_campaign)
        df = df[df["campaign"] != "other"].copy()
        df["innova_point"] = df.apply(assign_innova_point, axis=1)
        df = df[df["innova_point"] != "no"].copy()
        with open("csv/innova/notswitched/features_decomposition_list.json", "r") as f:
            dict_features = json.load(f)
    elif target_col[0] == "i_CO2":
        df["campaign"] = df["datetime"].apply(assign_campaign_iCO2)
        df = df[df["campaign"] != "other"].copy()
        df["innova_point"] = df.apply(assign_innova_point_iCO2, axis=1)
        df = df[df["innova_point"] != "no"].copy()
        with open("csv/innova/switched/features_decomposition_CO2_list.json", "r") as f:
            dict_features = json.load(f)
    else:
        raise ValueError("The target col do not match any existing function")

    df["hot"] = df.apply(assign_hot_season,axis=1)
    df["umidity_range"] = df.apply(assign_humidity, axis = 1)
    #Removing of rabbit 4
    df["column"] = df.apply(assign_colonna, axis = 1)
    df['height'] = df.apply(assign_altezza, axis = 1)

    return df,dict_features

```

```
def assign_innova_point(row):
    camp = row["campaign"]
    rid = row["id_rabbit"]

    # if camp == "2025_Jun":
    #     mapping = {2: "sl11", 4: "sl5", 6: "sl4", 1: "sl3", 5: "no"}
    # elif camp == "2025_Sep":
    #     mapping = {2: "sl11", 4: "sl5", 3: "sl4", 6: "sl3", 5: "no"}
    if camp == "2025_Apr":
        #mapping = {1: "no", 2: "no", 4: "no", 6: "sl5"}
        #mapping = {1: "no", 2: "sl5", 4: "no", 6: "sl5"}
        mapping = {1: "no", 2: "no", 4: "no", 6: "no"}
    elif camp == "2025_Jun":
        mapping = {2: "sl11", 4: "sl5", 1: "sl3", 6: "sl4", 5: "no"}
        #mapping = {2: "sl11", 4: "no", 1: "sl4", 6: "sl3", 5: "no"}
    elif camp == "2025_Sep":
        mapping = {2: "sl11", 4: "sl5", 6: "sl3", 3: "sl4", 5: "no"}
        #mapping = {2: "sl11", 4: "no", 6: "sl4", 3: "sl3", 5: "no"}
    elif camp == "2026_Gen":
        #mapping = {1: "no", 4: "no", 5: "sl11", 6: "no"}
        #mapping = {1: "sl11", 4: "no", 5: "sl11", 6: "sl11"}
        mapping = {1: "no", 4: "no", 5: "no", 6: "no"}
    else:
        mapping = {}

    return mapping.get(rid, "no")
```

Function print :

1.2 Metadata

Nombre de lignes df	8168
Nombre de lignes df clean	8168
Nombre de features	5
% de NaN	0.0
Taille dataset entraînement	5717
Taille dataset test	2450
Métrique utilisée	r2

Column target : i_NH3

Features_X : r_NH3, r_temperature, r_umidity, r_CO2, r_THI

feats_to_balance :

feats_grouping :

feats_postproc : datetime, id_channel, id_rabbit, campaign

feats_ref :

id_cols : campaign

2 Model(s) analysis

2.1 Explanation legend tables

Columns

- rank: ranking regarding cross validation test
- train_cv: mean score train of the CV
- test_cv: mean score test of the CV
- test: score on the testing dataset
- diff: test-test_cv
- r_test: test/test_cv
- r_train: train_cv/test_cv

Colors:

[Orange] $r_{\text{test}} < 0.95$ (under performance of the test)

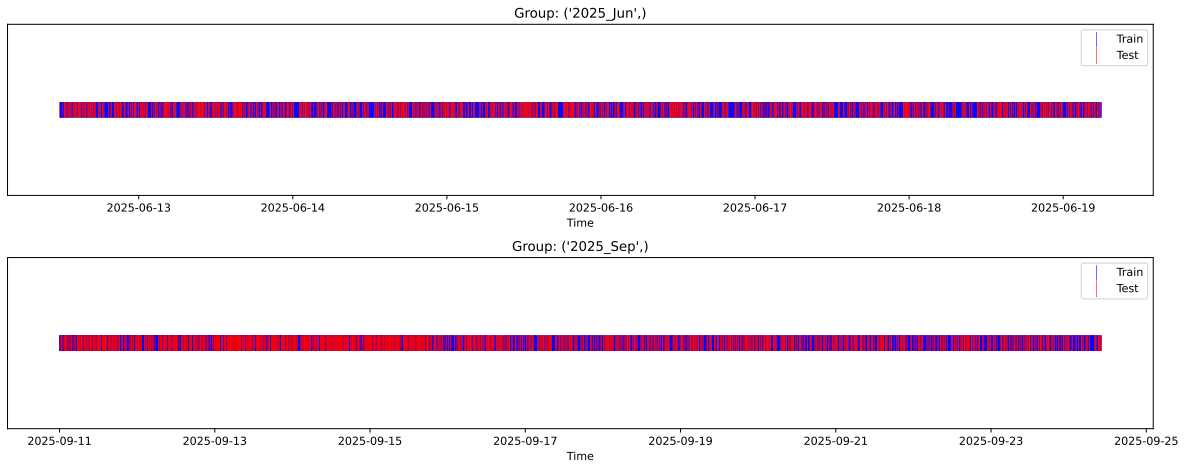
[Green] $r_{\text{test}} > 1.05$ (over performance of the test)

[Red] r_{train} not in $[0.95, 1.05]$ (gap between train/test)

2.2 Repartition of train and test

2.2.1 External Split: X_{train} vs X_{test}

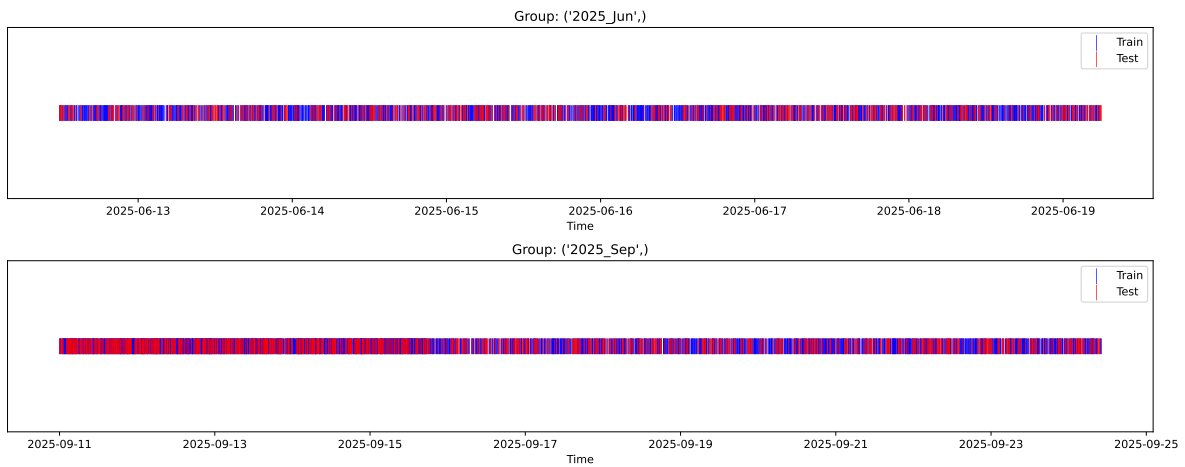
External Split: Train: 5717 samples Test: 2451 samples



2.2.2 Internal CV Folds

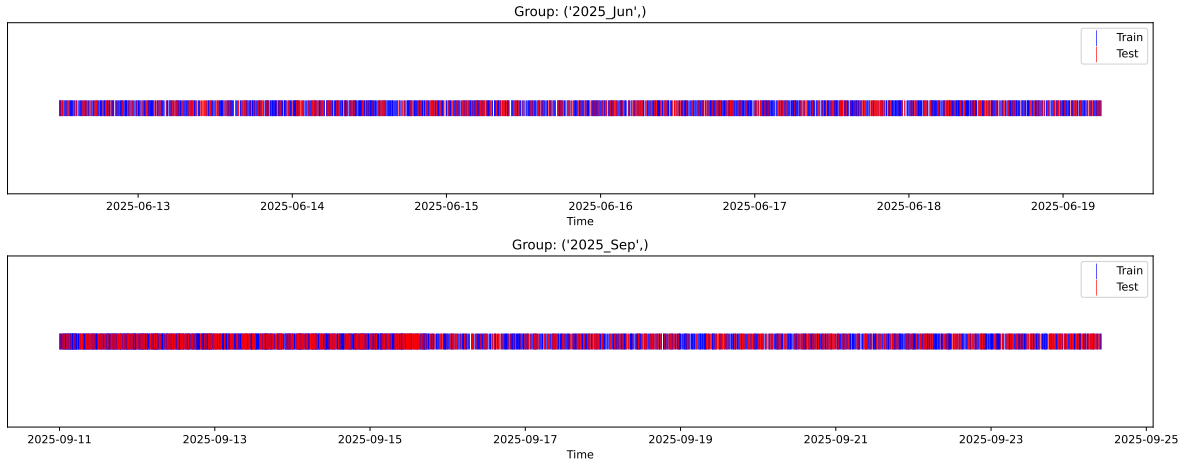
Fold 1/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



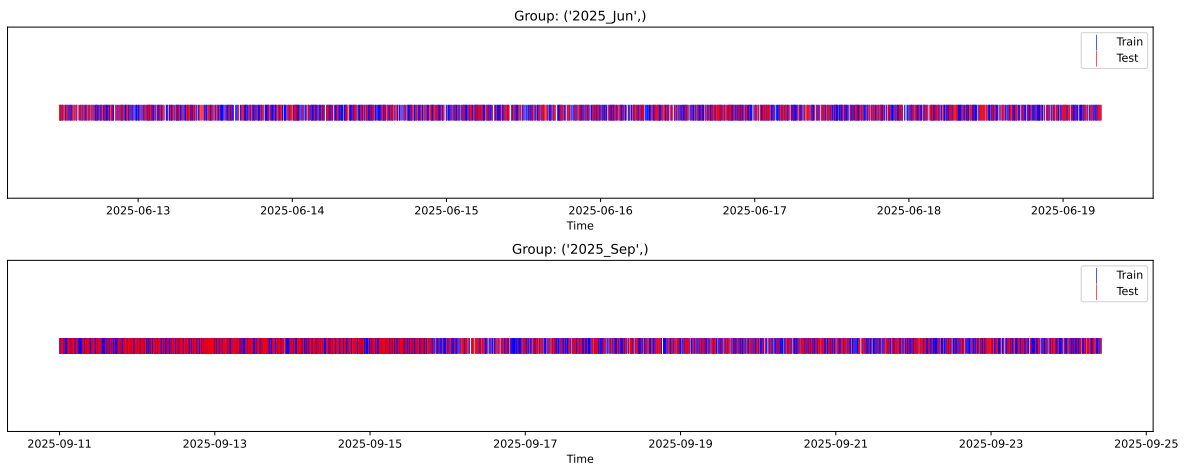
Fold 2/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



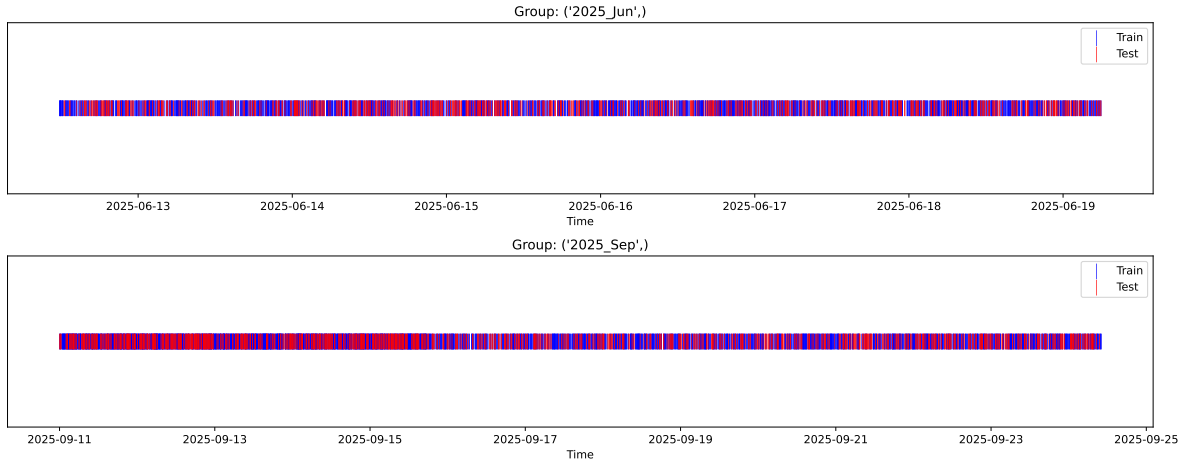
Fold 3/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



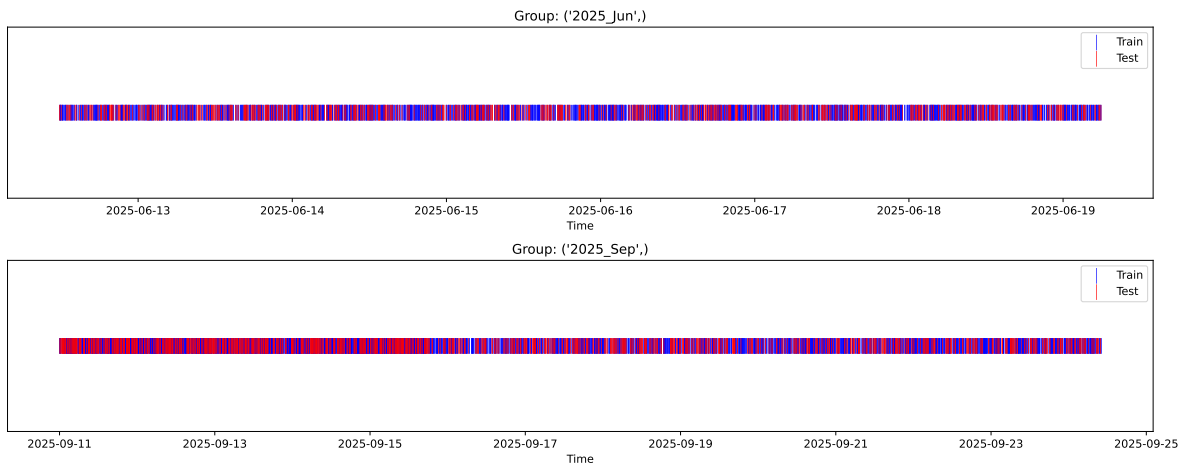
Fold 4/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



Fold 5/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h

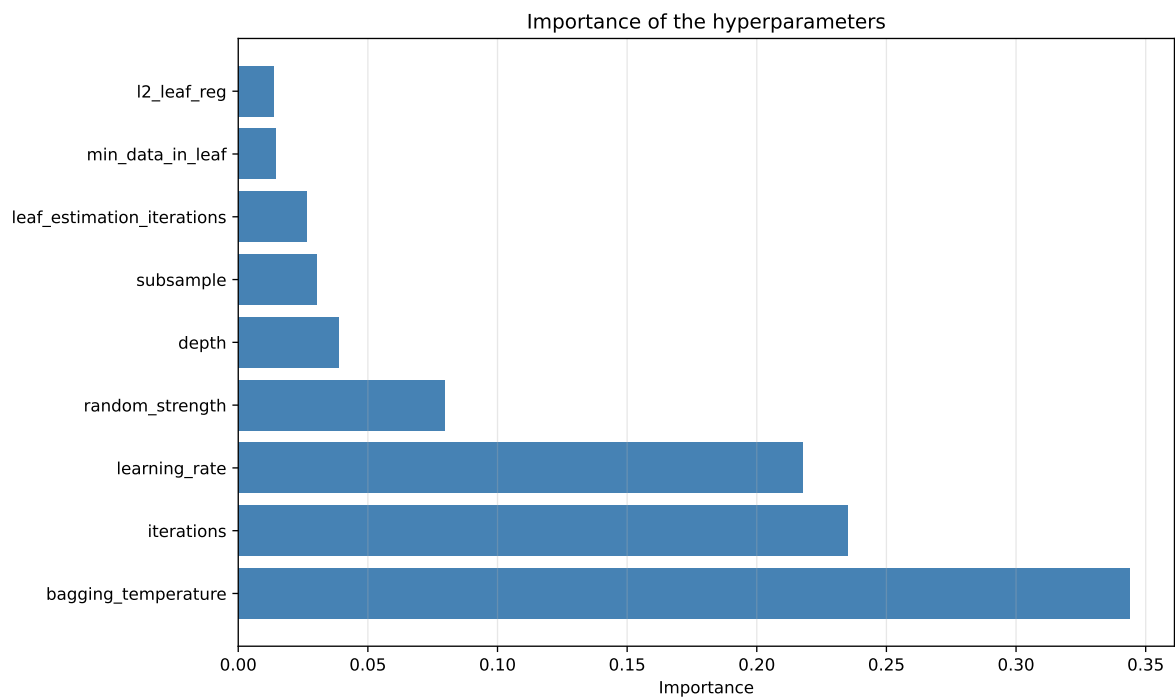
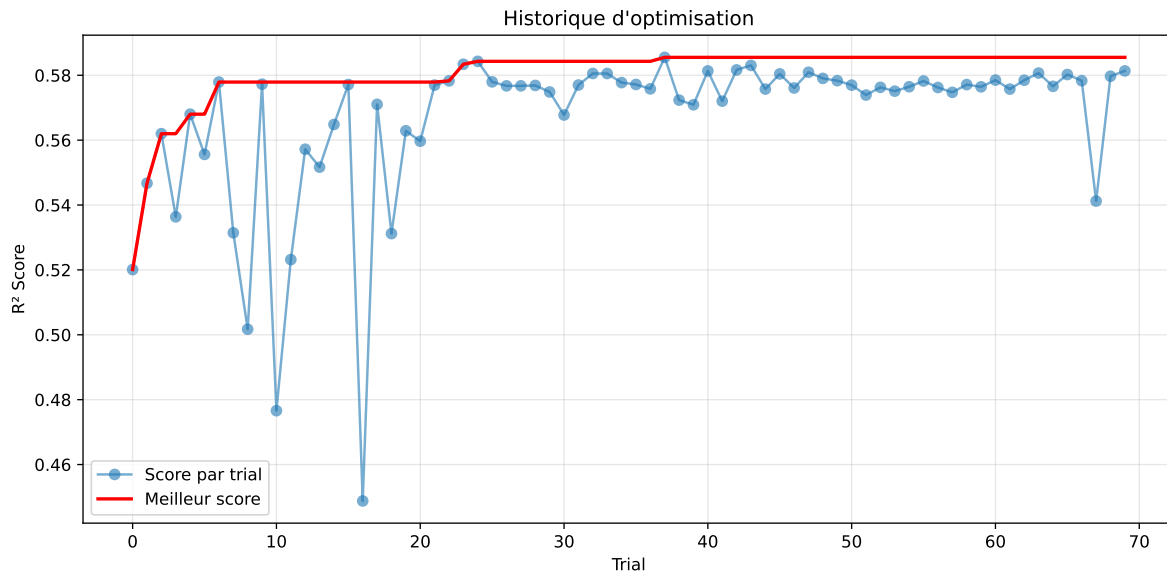


2.3 Model : CatBoost

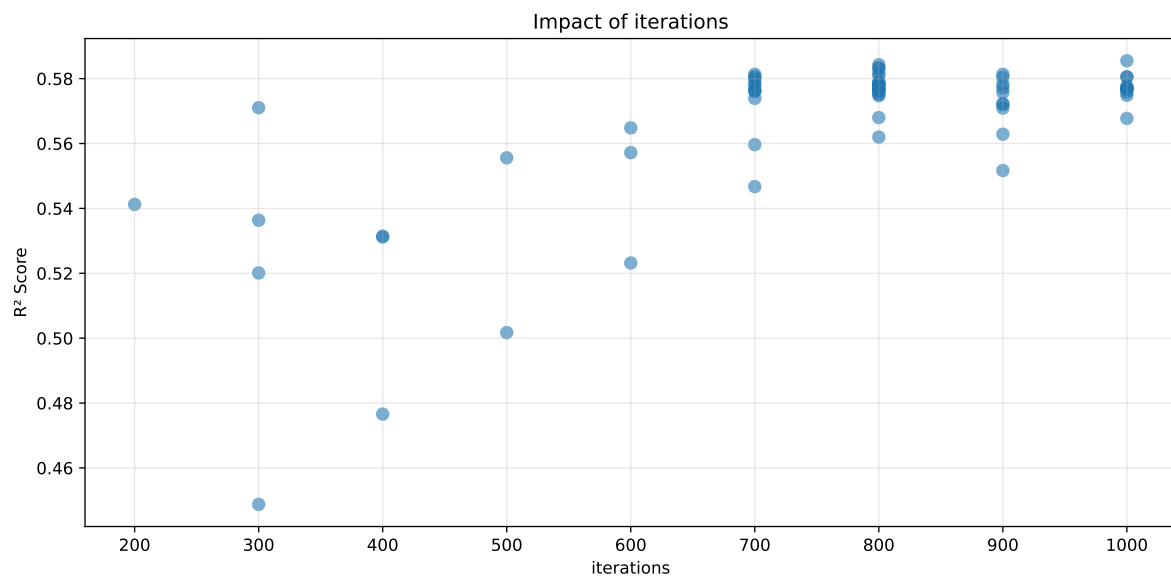
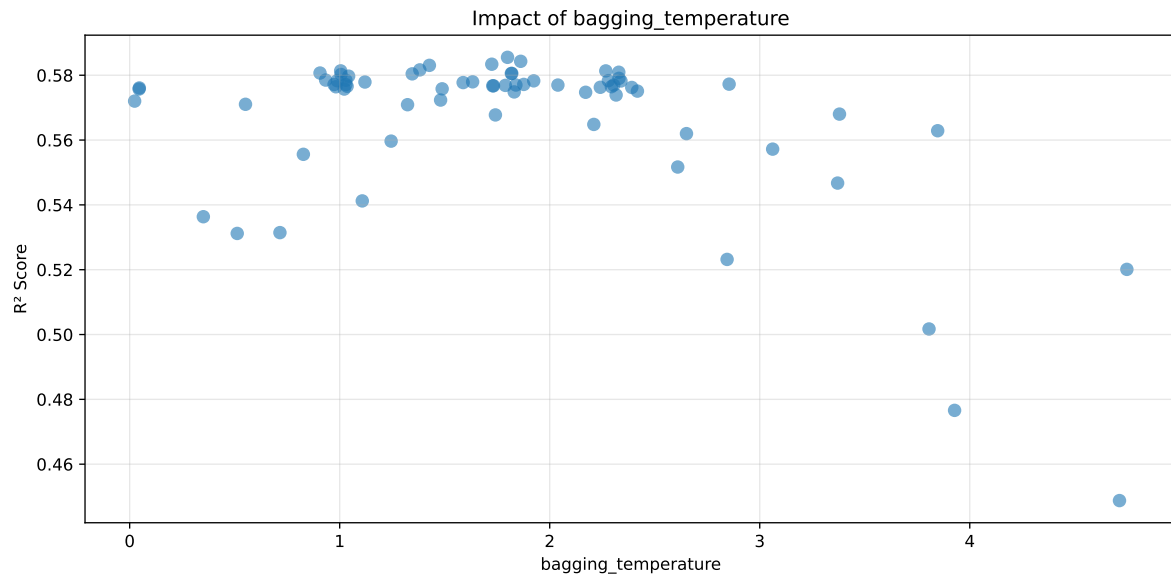
2.3.1 Gridsearch

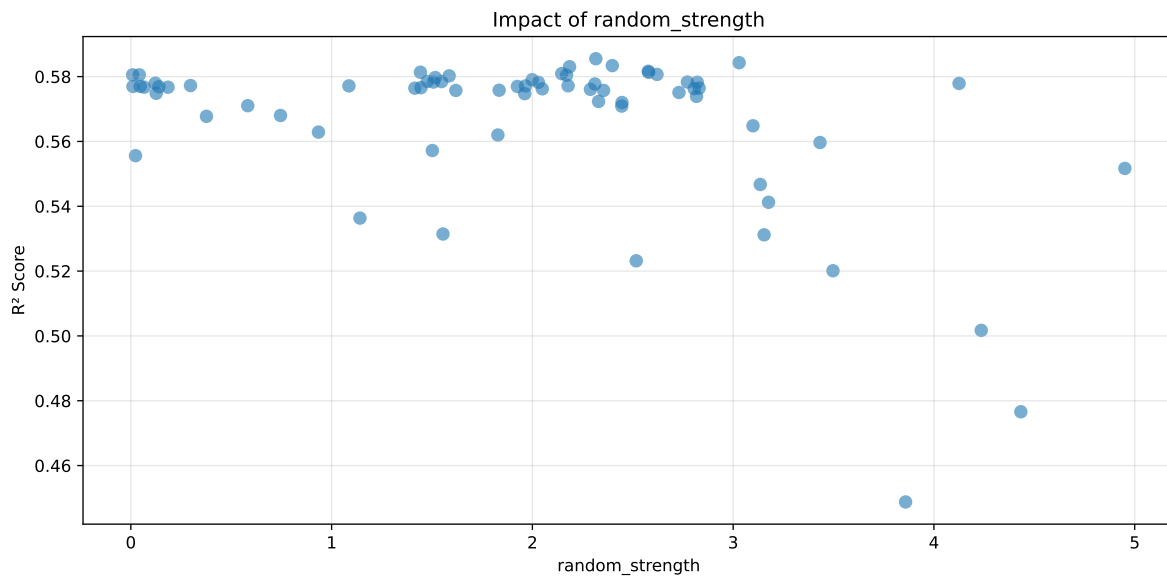
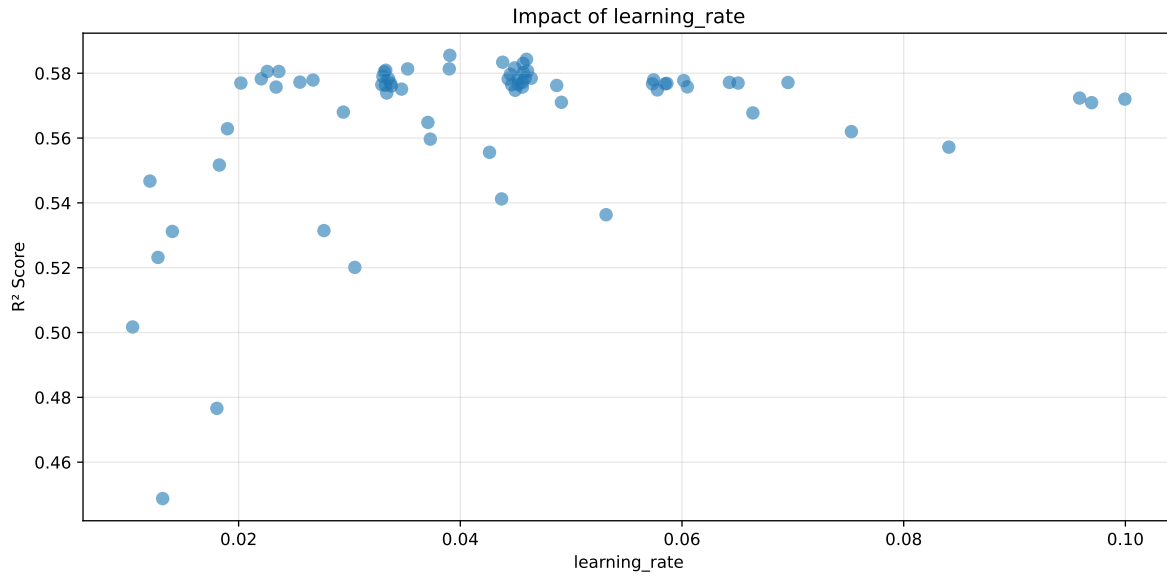
Distribution y_{train} vs y_{test} : y_{train} : mean=2.339, std=0.798, min=0.959, max=13.845
 y_{test} : mean=2.374, std=0.786, min=1.071, max=7.277

===== TOP Importance FEATURES =====
 feature importance $r_{umidity}$ 47.012423 r_{NH3} 16.589862 r_{CO2} 16.495906 r_{THI} 10.333629
 $r_{temperature}$ 9.568180



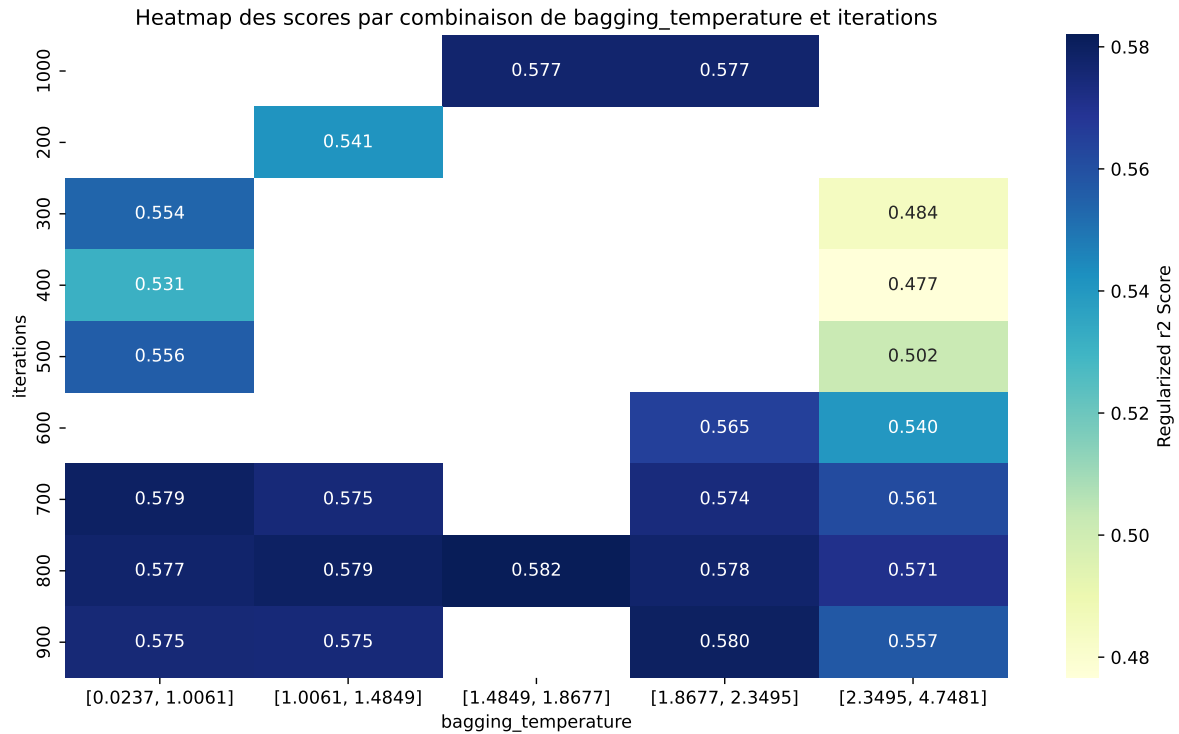
Paramètres avec >5% d'importance : ['bagging_temperature', 'iterations', 'learning_rate', 'random_strength']





C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.4935	0.4766	0.4963	0.0379	0.4766	0.507	0.0304	1.0639	1.0354
2	0.4593	0.4488	0.4594	0.0376	0.4488	0.4662	0.0175	1.039	1.0235
3	0.5524	0.5201	0.5465	0.038	0.4233	0.5535	0.0334	1.0642	1.0621
4	0.5305	0.5017	0.5233	0.0384	0.4152	0.532	0.0302	1.0603	1.0575
5	0.5639	0.5232	0.5509	0.0378	0.401	0.561	0.0378	1.0723	1.0779
6	0.5791	0.5314	0.5524	0.0366	0.3885	0.5564	0.0249	1.0469	1.0897
7	0.581	0.5312	0.5585	0.0378	0.3818	0.5679	0.0367	1.0691	1.0937
8	0.5934	0.5363	0.5666	0.0368	0.3653	0.5761	0.0398	1.0742	1.1063
9	0.6014	0.5412	0.5721	0.0374	0.3607	0.5769	0.0356	1.0658	1.1112
10	0.615	0.5467	0.5742	0.0368	0.3419	0.5812	0.0345	1.0631	1.1249
---	---	---	---	---	---	---	---	---	---
70	0.9611	0.5678	0.609	0.0362	-0.6122	0.6197	0.052	1.0916	1.6928

Hyperparameters:

Rank 1: {'reg__iterations': 400, 'reg__depth': 5, 'reg__learning_rate': 0.018040732438018663, 'reg__l2_leaf_reg': 48.97000769807214, 'reg__bagging_temperature': 3.9277390685489073, 'reg__random_strength': 4.432875452377819, 'reg__subsample': 0.5318111997080796, 'reg__min_data_in_leaf': 34, 'reg__leaf_estimation_iterations': 1}

Rank 2: {'reg__iterations': 300, 'reg__depth': 4, 'reg__learning_rate': 0.013147614024926053, 'reg__l2_leaf_reg': 41.743648700096585, 'reg__bagging_temperature': 4.712918000868085, 'reg__random_strength': 3.8588639744444, 'reg__subsample': 0.7196011443800487, 'reg__min_data_in_leaf': 44, 'reg__leaf_estimation_iterations': 8}

Rank 3: {'reg__iterations': 300, 'reg__depth': 5, 'reg__learning_rate': 0.030483488410241975, 'reg__l2_leaf_reg': 44.38897761331627, 'reg__bagging_temperature': 4.748089959636791, 'reg__random_strength': 3.4971500660609536, 'reg__subsample': 0.5694164430312532, 'reg__min_data_in_leaf': 42, 'reg__leaf_estimation_iterations': 5}

Rank 4: {'reg__iterations': 500, 'reg__depth': 6, 'reg__learning_rate': 0.010439641555251952, 'reg__l2_leaf_reg': 20.125465690535105, 'reg__bagging_temperature': 3.8060509634440107, 'reg__random_strength': 4.2356731700655255, 'reg__subsample': 0.594652039074104, 'reg__min_data_in_leaf': 26, 'reg__leaf_estimation_iterations': 6}

Rank 5: {'reg__iterations': 600, 'reg__depth': 7, 'reg__learning_rate': 0.012727380597685668, 'reg__l2_leaf_reg': 22.32326594393304, 'reg__bagging_temperature': 2.8446078693482906, 'reg__random_strength': 2.517258664137655, 'reg__subsample': 0.6276534902048069, 'reg__min_data_in_leaf': 45, 'reg__leaf_estimation_iterations': 1}

Rank 6: {'reg__iterations': 400, 'reg__depth': 4, 'reg__learning_rate': 0.0276985158136313, 'reg__l2_leaf_reg': 7.120109142544764, 'reg__bagging_temperature': 0.7154847750903837, 'reg__random_strength': 1.5547217862515121, 'reg__subsample': 0.5471165252712364, 'reg__min_data_in_leaf': 36, 'reg__leaf_estimation_iterations': 3}

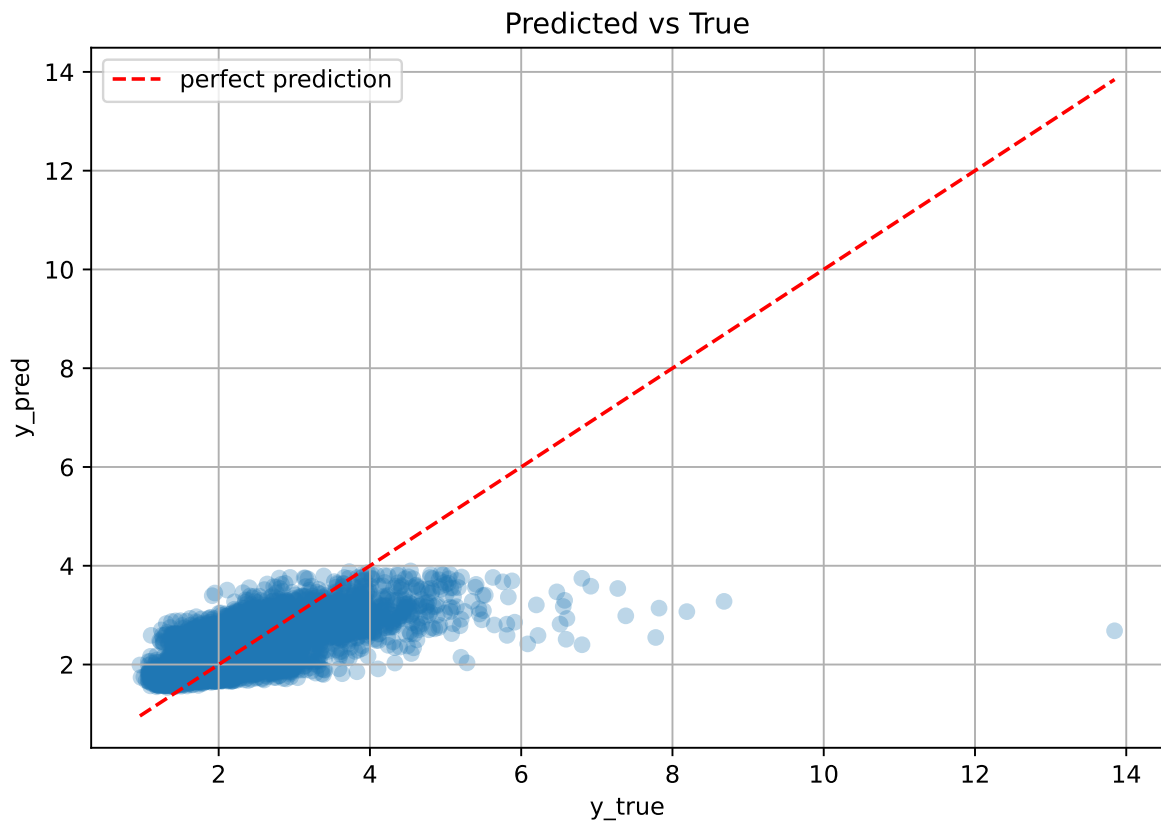
Rank 7: {'reg__iterations': 400, 'reg__depth': 8, 'reg__learning_rate': 0.014025273459606906, 'reg__l2_leaf_reg': 34.3139480038702, 'reg__bagging_temperature': 0.5121290678612944, 'reg__random_strength': 3.154189575725489, 'reg__subsample': 0.8196778455929946, 'reg__min_data_in_leaf': 45, 'reg__leaf_estimation_iterations': 6}

Rank 8: {'reg__iterations': 300, 'reg__depth': 6, 'reg__learning_rate': 0.053151624153126145, 'reg__l2_leaf_reg': 49.72635996263596, 'reg__bagging_temperature': 0.35079245877275955, 'reg__random_strength': 1.1411777016701157, 'reg__subsample': 0.5393174055343588, 'reg__min_data_in_leaf': 21, 'reg__leaf_estimation_iterations': 1}

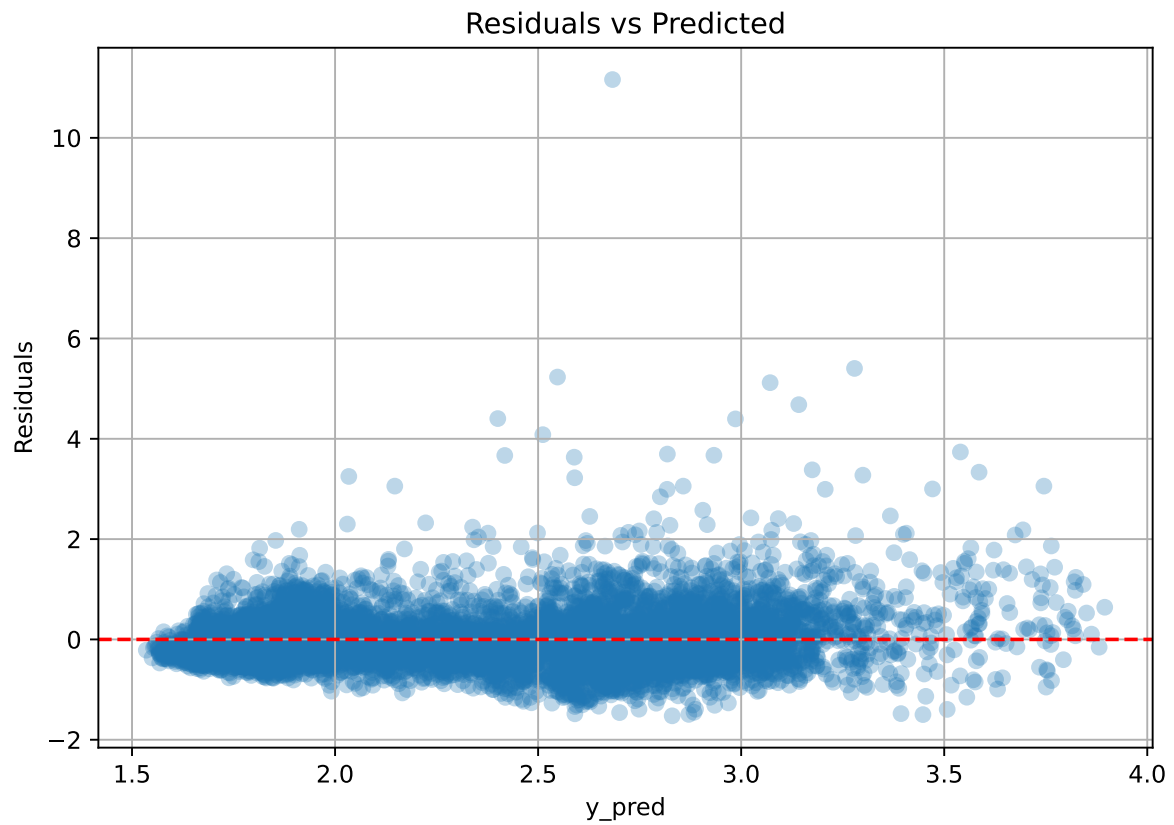
Rank 9: {'reg__iterations': 200, 'reg__depth': 7, 'reg__learning_rate': 0.04372560129698771, 'reg__l2_leaf_reg': 42.728193807433804, 'reg__bagging_temperature': 1.1075238432481698, 'reg__random_strength': 3.1762273530328153, 'reg__subsample': 0.6985502806385782, 'reg__min_data_in_leaf': 14, 'reg__leaf_estimation_iterations': 10}

Rank 10: {'reg__iterations': 700, 'reg__depth': 7, 'reg__learning_rate': 0.011998491306351902, 'reg__l2_leaf_reg': 7.883776541398703, 'reg__bagging_temperature': 3.37082387307786, 'reg__random_strength': 3.1349514083245253, 'reg__subsample': 0.7899588072256345, 'reg__min_data_in_leaf': 21, 'reg__leaf_estimation_iterations': 6}

2.3.2 Scatter



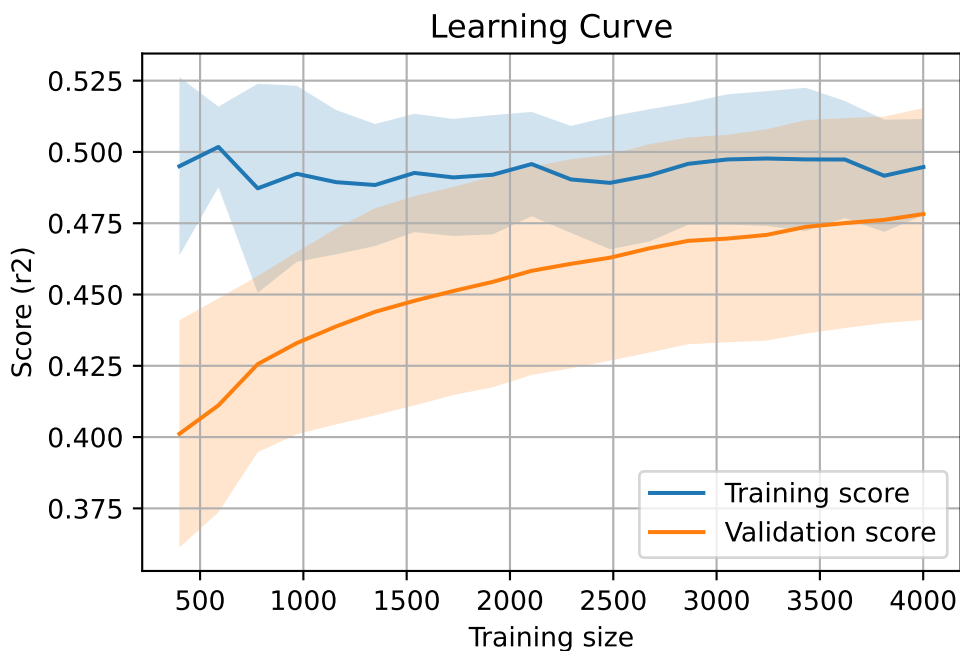
2.3.3 Scatter residuals



2.3.4 Learning curve

Len(group) : 8168, **Len(training)** : 5718,

Len(training_real) : 4003, **Len(test_of_training)** : 1715, **Len(test)** : 2450



2.4 Model : LGBM

2.4.1 Gridsearch

Distribution y_{train} vs y_{test} : y_{train} : mean=2.339, std=0.798, min=0.959, max=13.845
 y_{test} : mean=2.374, std=0.786, min=1.071, max=7.277

```
C:\Users\titou\anaconda3\Lib\site-packages\sklearn\utils\validation.py:2749: UserWarning:
X does not have valid feature names, but LGBMRegressor was fitted with feature names
C:\Users\titou\anaconda3\Lib\site-packages\sklearn\utils\validation.py:2749: UserWarning:
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[illegible]

[illegible]

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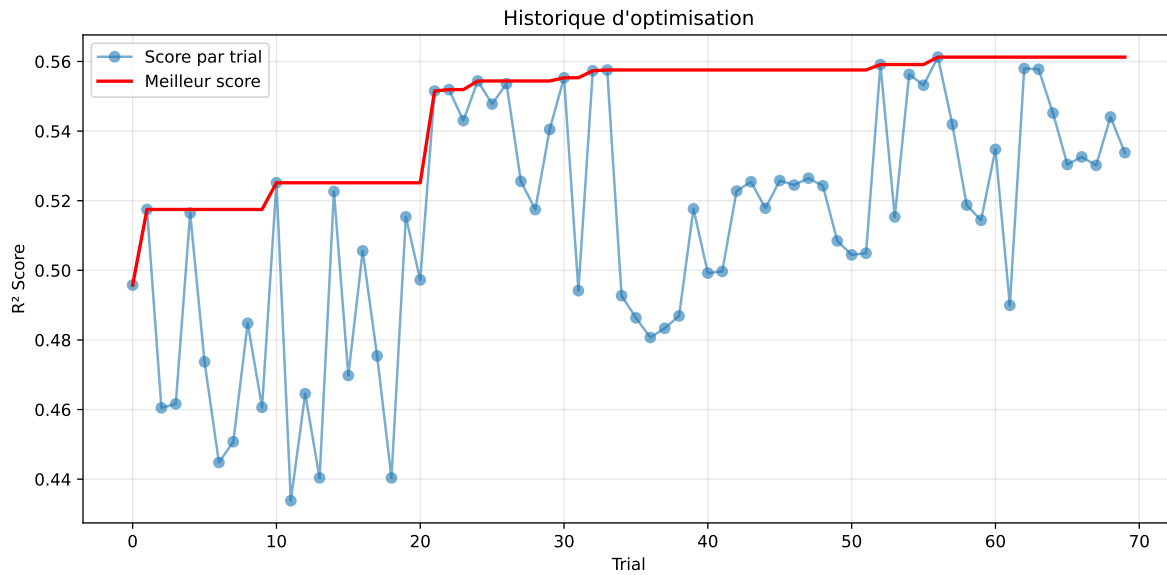
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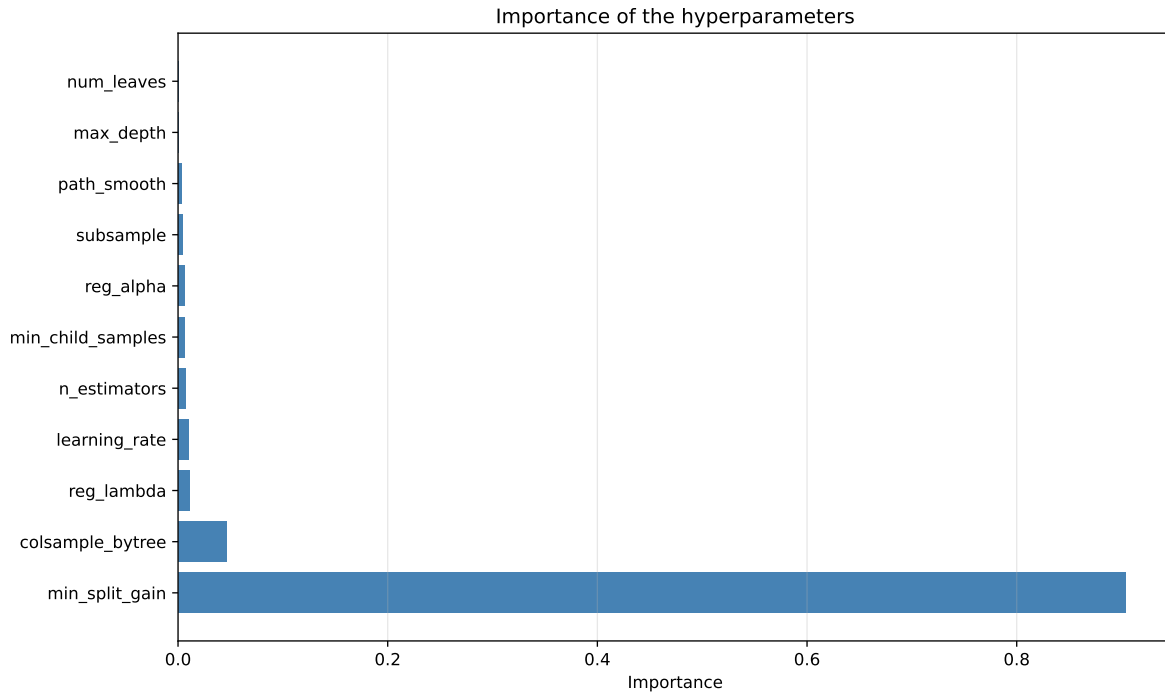
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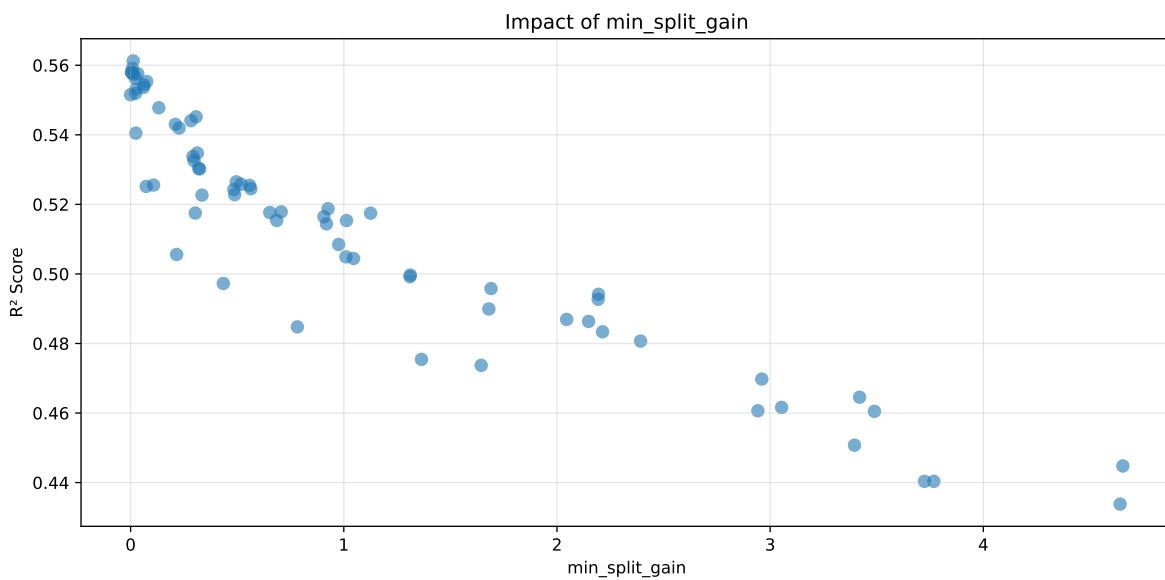
===== TOP Importance FEATURES =====

feature importance r_CO2 176 r_umidity 158 r_NH3 119 r_THI 84 r_temperature 77





Paramètres avec >5% d'importance : ['min_split_gain']



Pas assez de paramètres avec >5% d'importance pour une heatmap 2D.

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rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.4723	0.4508	0.4637	0.0334	0.4508	0.4899	0.0392	1.0869	1.0478
2	0.462	0.4404	0.4512	0.0355	0.4404	0.4791	0.0388	1.0881	1.0491
3	0.4598	0.4404	0.4519	0.0374	0.4404	0.4787	0.0383	1.087	1.0441
4	0.5164	0.4848	0.5047	0.0358	0.3899	0.5215	0.0367	1.0757	1.0652
5	0.4869	0.4616	0.4773	0.0348	0.3857	0.4964	0.0348	1.0753	1.0548
6	0.5182	0.4834	0.5041	0.0377	0.3788	0.5192	0.0358	1.0741	1.0721
7	0.5056	0.4737	0.491	0.0361	0.378	0.5103	0.0365	1.0771	1.0673
8	0.523	0.4863	0.5082	0.0397	0.3764	0.5273	0.0409	1.0842	1.0754
9	0.5239	0.4869	0.5084	0.0382	0.3761	0.53	0.0431	1.0886	1.0759
10	0.5157	0.4807	0.5032	0.0392	0.3756	0.524	0.0433	1.0901	1.0729
---	---	---	---	---	---	---	---	---	---
70	0.8345	0.5515	0.5948	0.0362	-0.2973	0.6116	0.0601	1.1089	1.513

Hyperparameters:

Rank 1: {'reg__n_estimators': 700, 'reg__max_depth': 3, 'reg__learning_rate': 0.038701506879767665, 'reg__num_leaves': 50, 'reg__subsample': 0.800445048887491, 'reg__colsample_bytree': 0.5909189620188818, 'reg__min_child_samples': 48, 'reg__reg_alpha': 2.8637527517705443, 'reg__reg_lambda': 16.53375972772102, 'reg__min_split_gain': 3.395699855475629, 'reg__path_smooth': 2.087976899759595}

Rank 2: {'reg__n_estimators': 500, 'reg__max_depth': 8, 'reg__learning_rate': 0.030127410318227195, 'reg__num_leaves': 28, 'reg__subsample': 0.825941733750851, 'reg__colsample_bytree': 0.5536380182393499, 'reg__min_child_samples': 31, 'reg__reg_alpha': 5.711416630452051, 'reg__reg_lambda': 43.852183761746616, 'reg__min_split_gain': 3.72395868142672, 'reg__path_smooth': 3.0974032083820187}

Rank 3: {'reg__n_estimators': 300, 'reg__max_depth': 3, 'reg__learning_rate': 0.032069209611523104, 'reg__num_leaves': 53, 'reg__subsample': 0.8438859697858969, 'reg__colsample_bytree': 0.6748451474404602, 'reg__min_child_samples': 40, 'reg__reg_alpha': 6.6688010604282955, 'reg__reg_lambda': 5.64787655005844, 'reg__min_split_gain': 3.767722498019711, 'reg__path_smooth': 0.3808947802468121}

Rank 4: {'reg__n_estimators': 600, 'reg__max_depth': 3, 'reg__learning_rate': 0.01666115588981105, 'reg__num_leaves': 18, 'reg__subsample': 0.6967735340795551, 'reg__colsample_bytree': 0.7959485143144829, 'reg__min_child_samples': 47, 'reg__reg_alpha': 6.150048267982777, 'reg__reg_lambda': 37.526646439429676, 'reg__min_split_gain': 0.7823497031019355, 'reg__path_smooth': 1.9904556257619987}

Rank 5: {'reg__n_estimators': 200, 'reg__max_depth': 3, 'reg__learning_rate': 0.037629696657324435, 'reg__num_leaves': 59, 'reg__subsample': 0.5594143818244706,

'reg__colsample_bytree': 0.5323390807470206, 'reg__min_child_samples': 33, 'reg__reg_alpha': 2.2844984573282314, 'reg__reg_lambda': 0.38244730933793547, 'reg__min_split_gain': 3.0538413264106197, 'reg__path_smooth': 1.088576598011926}

Rank 6: {'reg__n_estimators': 600, 'reg__max_depth': 4, 'reg__learning_rate': 0.09456716564808629, 'reg__num_leaves': 42, 'reg__subsample': 0.6064336371333257, 'reg__colsample_bytree': 0.8980726104703273, 'reg__min_child_samples': 23, 'reg__reg_alpha': 1.053707305079877, 'reg__reg_lambda': 29.70563864391815, 'reg__min_split_gain': 2.2138482351195066, 'reg__path_smooth': 4.9492524059968295}

Rank 7: {'reg__n_estimators': 500, 'reg__max_depth': 4, 'reg__learning_rate': 0.09855477480813987, 'reg__num_leaves': 28, 'reg__subsample': 0.7839165183118133, 'reg__colsample_bytree': 0.6103070511528206, 'reg__min_child_samples': 33, 'reg__reg_alpha': 6.881534152196302, 'reg__reg_lambda': 28.7234309333771, 'reg__min_split_gain': 1.6446368676705108, 'reg__path_smooth': 0.26335895100921325}

Rank 8: {'reg__n_estimators': 600, 'reg__max_depth': 4, 'reg__learning_rate': 0.04715084112568352, 'reg__num_leaves': 32, 'reg__subsample': 0.7685058039420662, 'reg__colsample_bytree': 0.8980163532406201, 'reg__min_child_samples': 23, 'reg__reg_alpha': 0.5502593709369042, 'reg__reg_lambda': 21.10286911069686, 'reg__min_split_gain': 2.148168781679286, 'reg__path_smooth': 4.843915127915332}

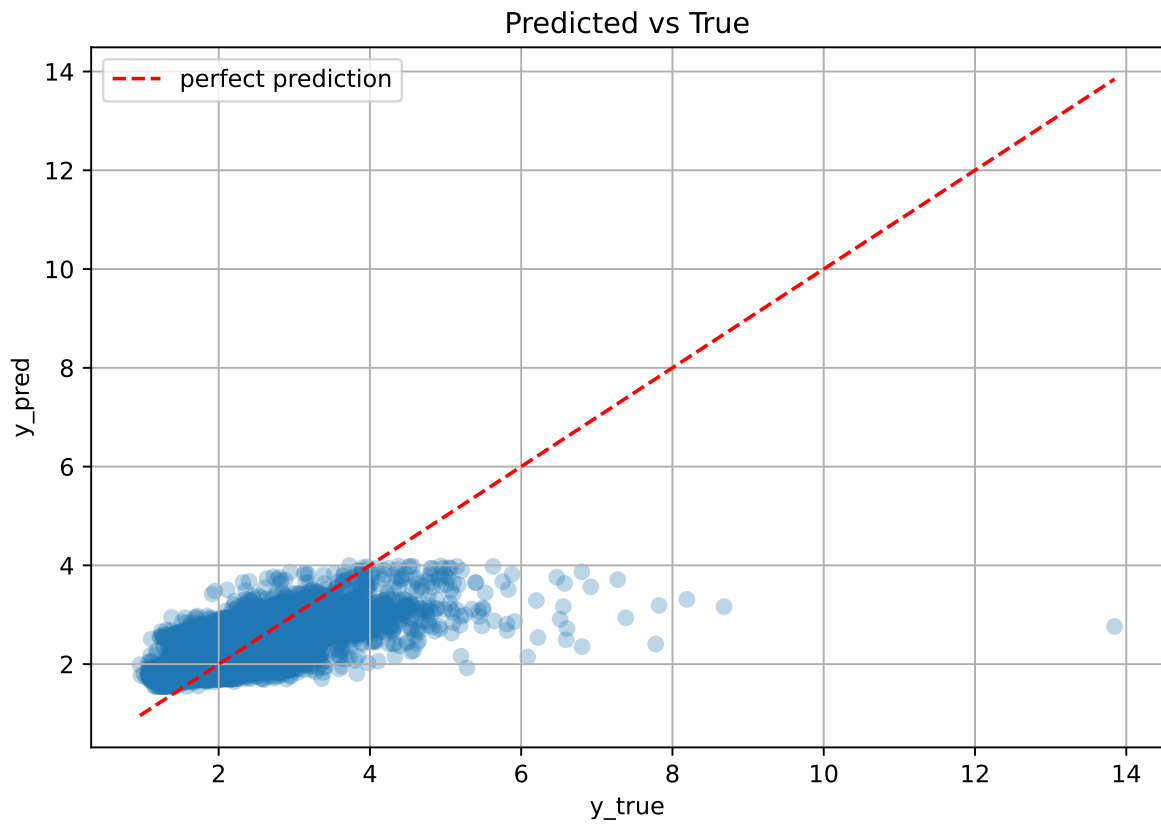
Rank 9: {'reg__n_estimators': 700, 'reg__max_depth': 4, 'reg__learning_rate': 0.08008811559557395, 'reg__num_leaves': 43, 'reg__subsample': 0.6647864789190302, 'reg__colsample_bytree': 0.8951226792899577, 'reg__min_child_samples': 21, 'reg__reg_alpha': 0.8757375187712123, 'reg__reg_lambda': 28.01862962241841, 'reg__min_split_gain': 2.045301149058275, 'reg__path_smooth': 4.474043217485255}

Rank 10: {'reg__n_estimators': 600, 'reg__max_depth': 4, 'reg__learning_rate': 0.05007844955755347, 'reg__num_leaves': 36, 'reg__subsample': 0.7454828913531582, 'reg__colsample_bytree': 0.8878462515277962, 'reg__min_child_samples': 23, 'reg__reg_alpha': 0.0643216930499216, 'reg__reg_lambda': 27.48827625378082, 'reg__min_split_gain': 2.392266658090709, 'reg__path_smooth': 4.977788841039627}

2.4.2 Scatter

C:\Users\titou\anaconda3\Lib\site-packages\sklearn\utils\validation.py:2749: UserWarning:

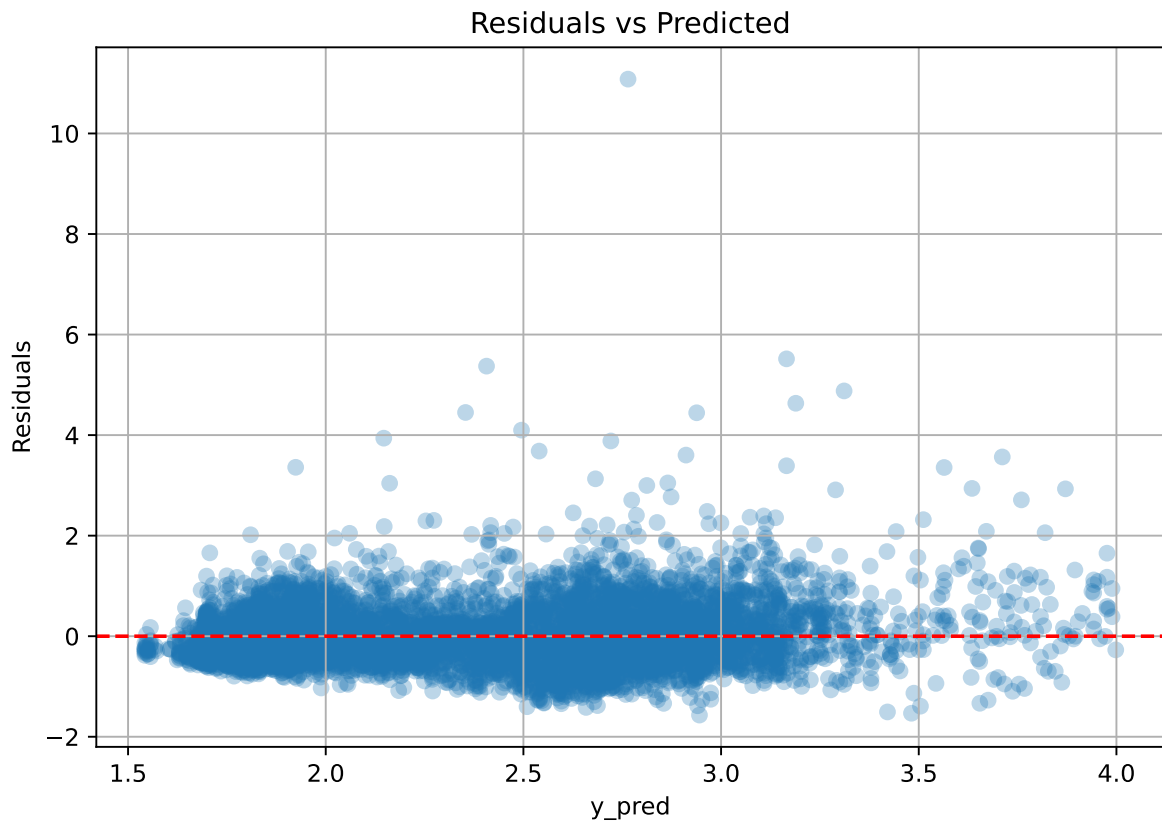
X does not have valid feature names, but LGBMRegressor was fitted with feature names



2.4.3 Scatter residuals

C:\Users\titou\anaconda3\Lib\site-packages\sklearn\utils\validation.py:2749: UserWarning:

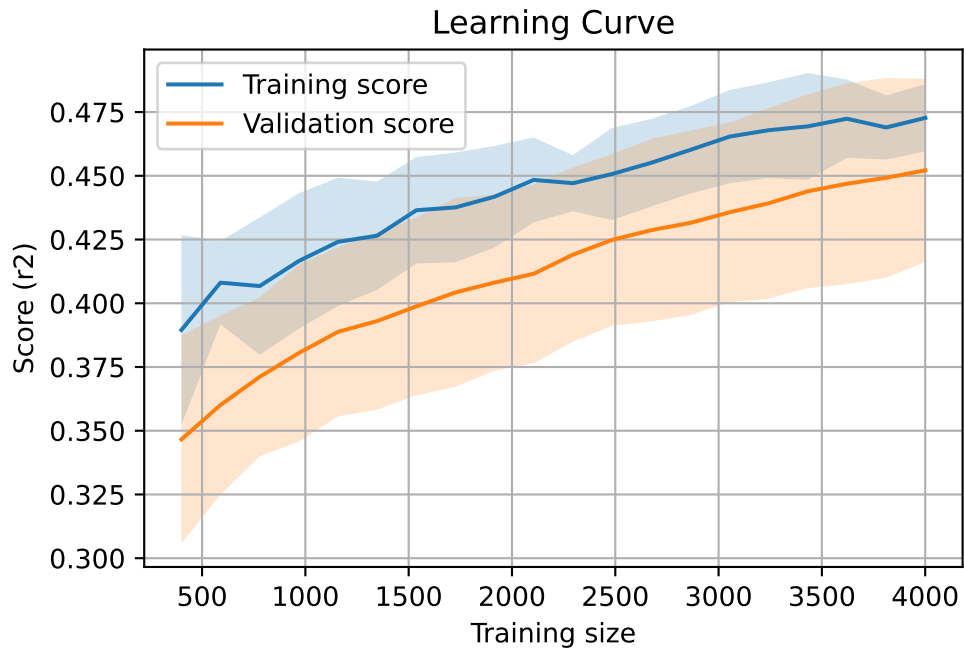
X does not have valid feature names, but LGBMRegressor was fitted with feature names



2.4.4 Learning curve

Len(group) : 8168, **Len(training)** : 5718,

Len(training_real) : 4003, **Len(test_of_training)** : 1715, **Len(test)** : 2450



2.5 Sumup-table

model	mean_train_cv	mean_test_cv	test_cv	std_test_cv	pen_score	test	diff	r_test	r_train
CatBoost	0.494	0.477	0.496	0.038	0.4766	0.507	0.03	1.064	1.035
LGBM	0.472	0.451	0.464	0.033	0.4508	0.49	0.039	1.087	1.048

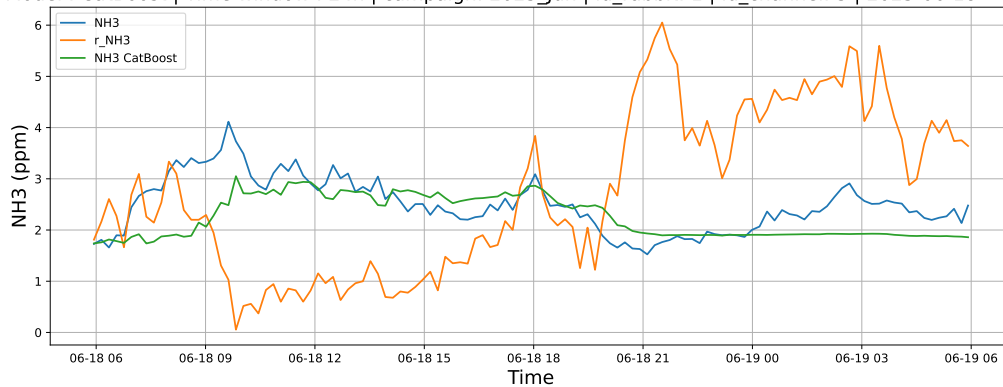
3 Plot

3.1 Standalone plots

3.1.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

3.1.1.1 Time window : 24

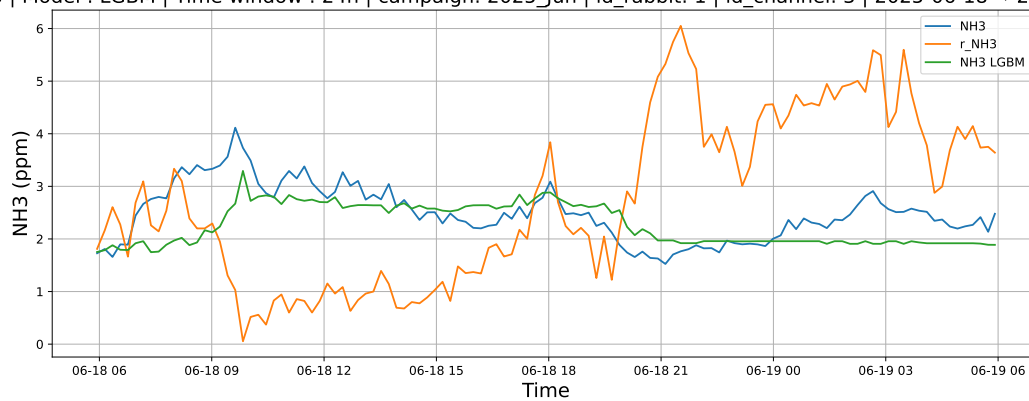
NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



C:\Users\titou\anaconda3\Lib\site-packages\sklearn\utils\validation.py:2749: UserWarning:

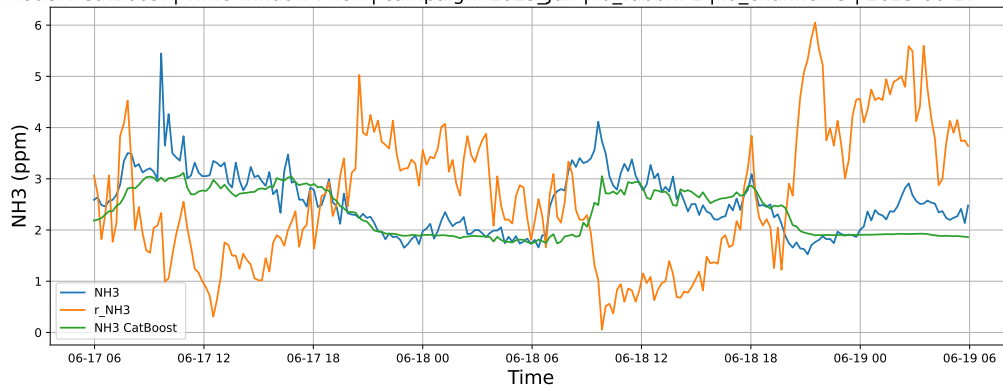
X does not have valid feature names, but LGBMRegressor was fitted with feature names

NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



3.1.1.2 Time window : 48

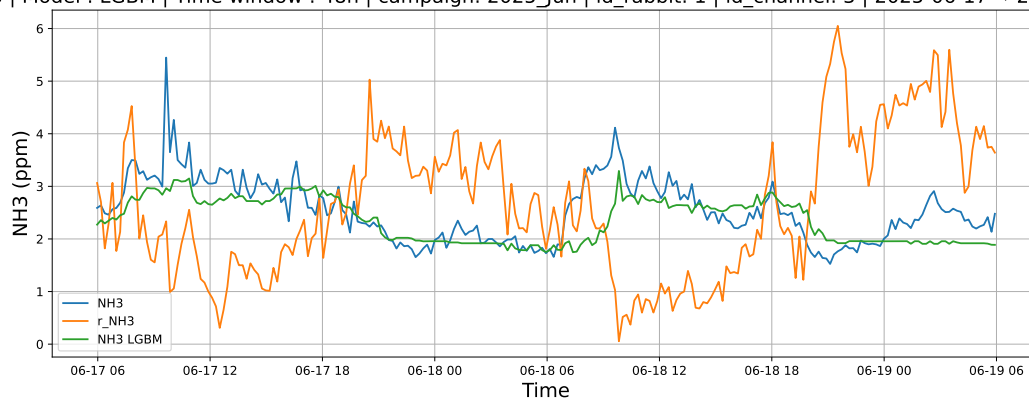
NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



C:\Users\titou\anaconda3\Lib\site-packages\sklearn\utils\validation.py:2749: UserWarning:

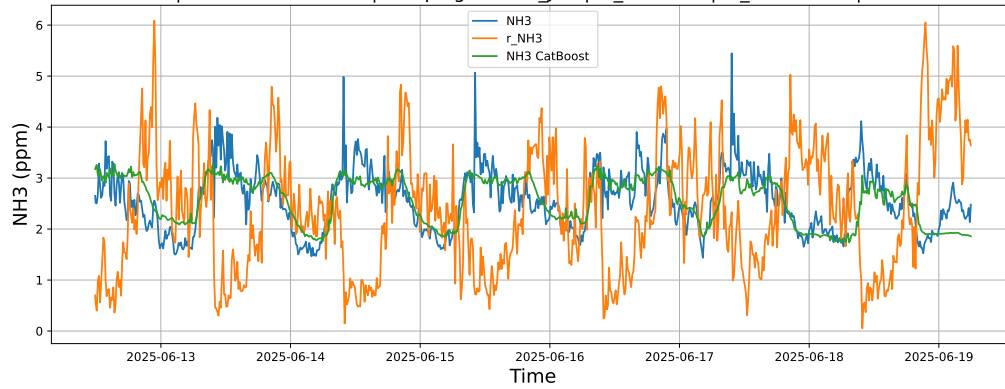
X does not have valid feature names, but LGBMRegressor was fitted with feature names

NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



3.1.1.3 Time window : ALL

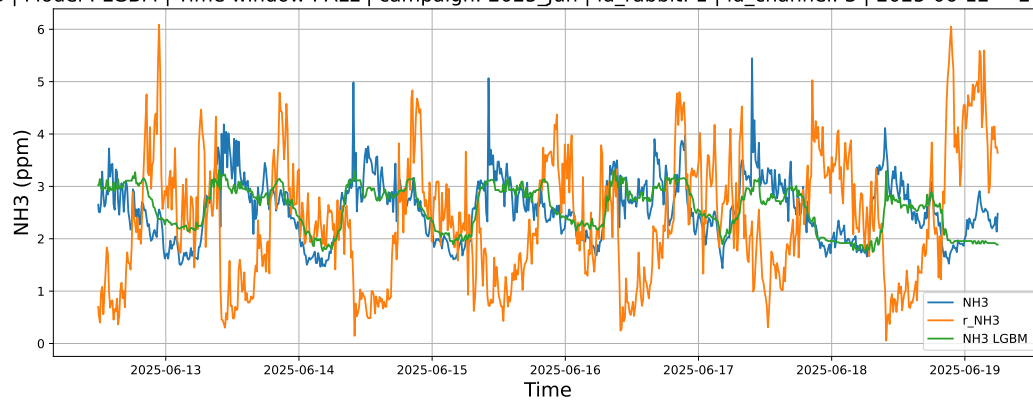
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



C:\Users\titou\anaconda3\Lib\site-packages\sklearn\utils\validation.py:2749: UserWarning:

X does not have valid feature names, but LGBMRegressor was fitted with feature names

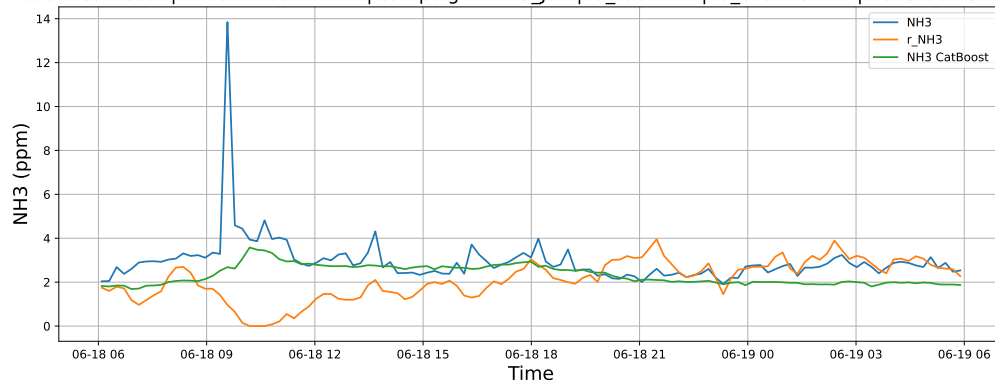
NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



3.1.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

3.1.2.1 Time window : 24

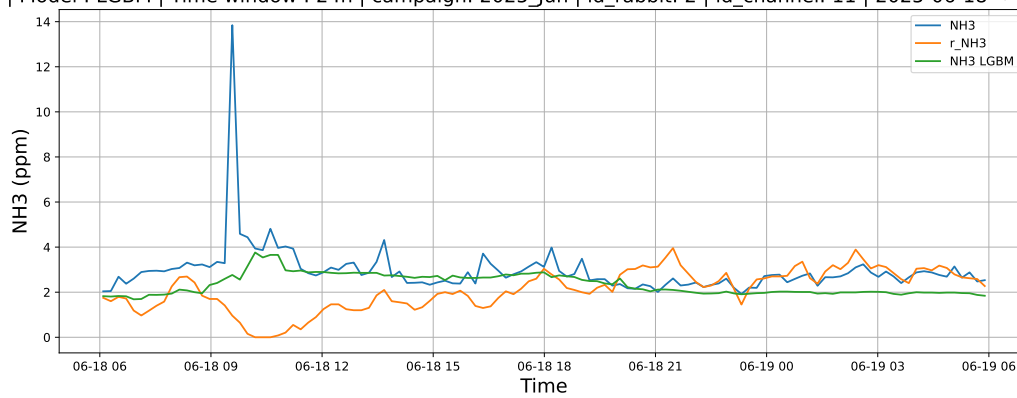
NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



C:\Users\titou\anaconda3\Lib\site-packages\sklearn\utils\validation.py:2749: UserWarning:

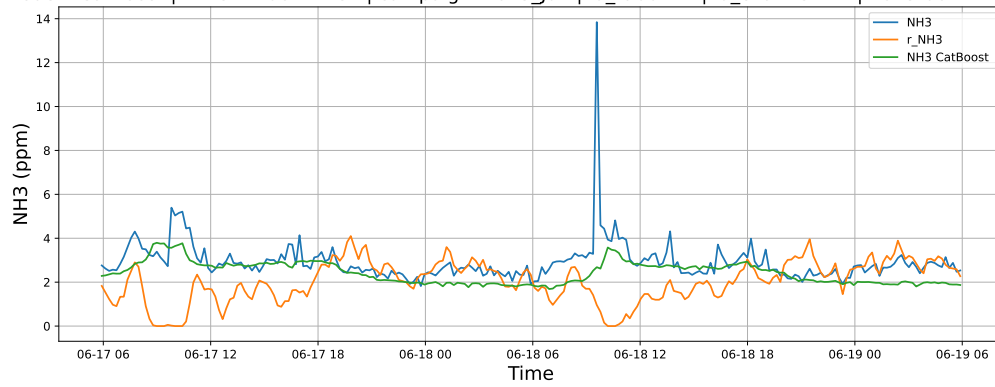
X does not have valid feature names, but LGBMRegressor was fitted with feature names

NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



3.1.2.2 Time window : 48

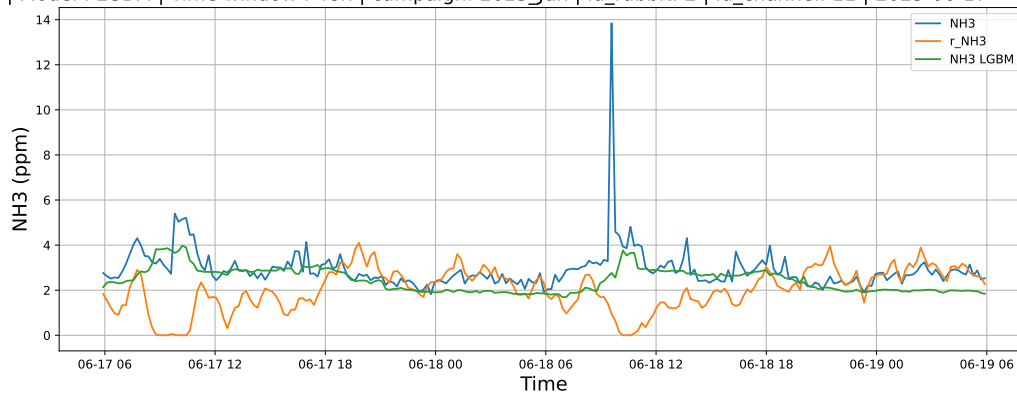
NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



C:\Users\titou\anaconda3\Lib\site-packages\sklearn\utils\validation.py:2749: UserWarning:

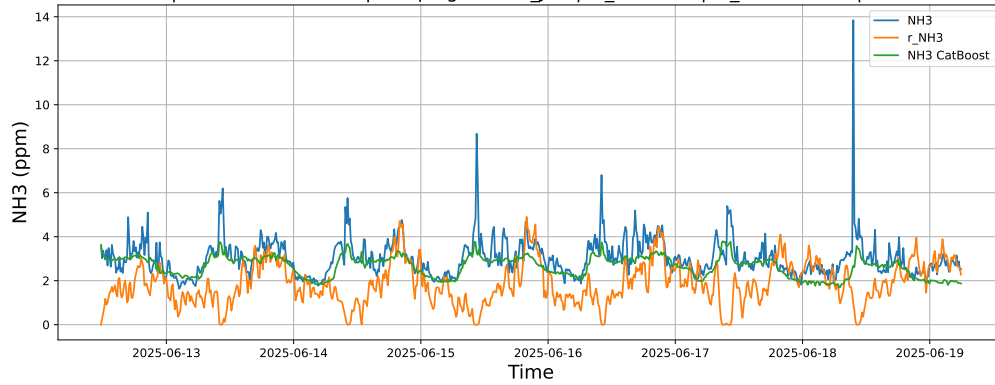
X does not have valid feature names, but LGBMRegressor was fitted with feature names

NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



3.1.2.3 Time window : ALL

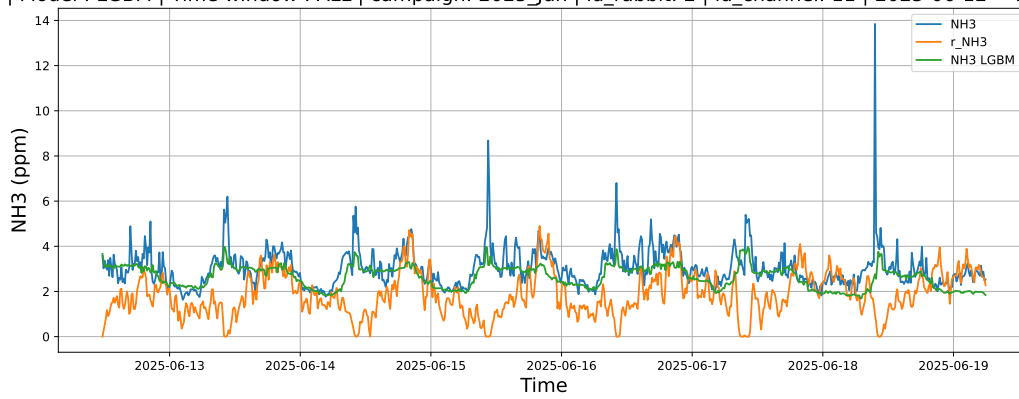
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



C:\Users\titou\anaconda3\Lib\site-packages\sklearn\utils\validation.py:2749: UserWarning:

X does not have valid feature names, but LGBMRegressor was fitted with feature names

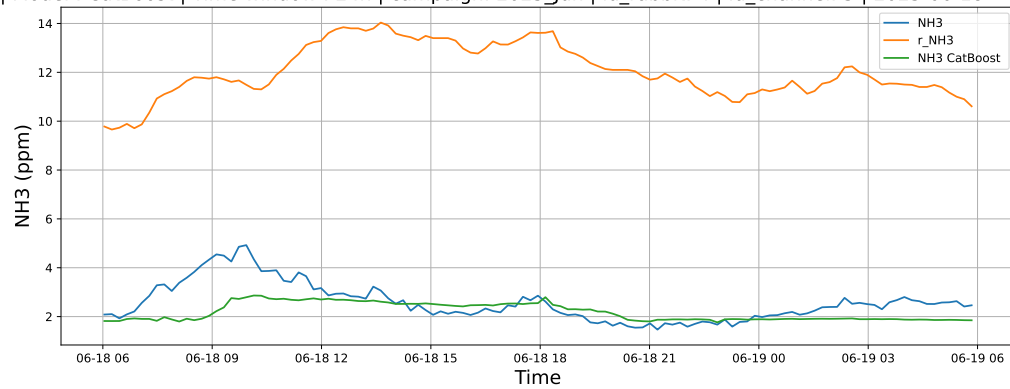
NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



3.1.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

3.1.3.1 Time window : 24

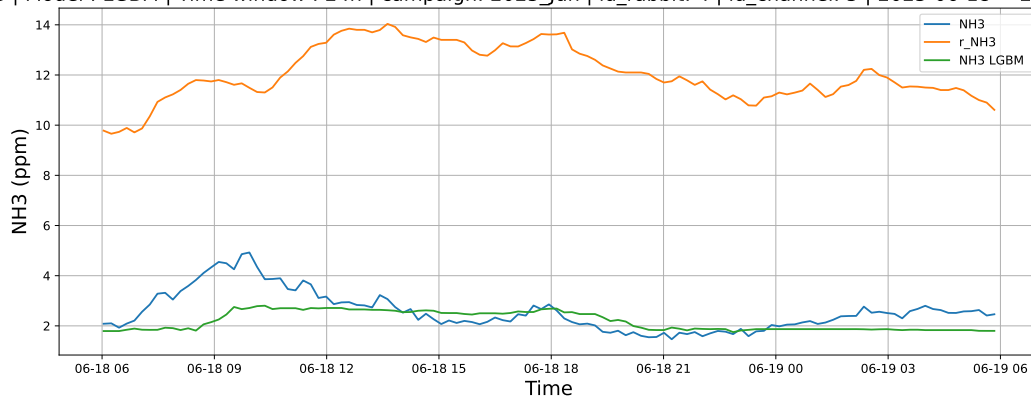
NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



C:\Users\titou\anaconda3\Lib\site-packages\sklearn\utils\validation.py:2749: UserWarning:

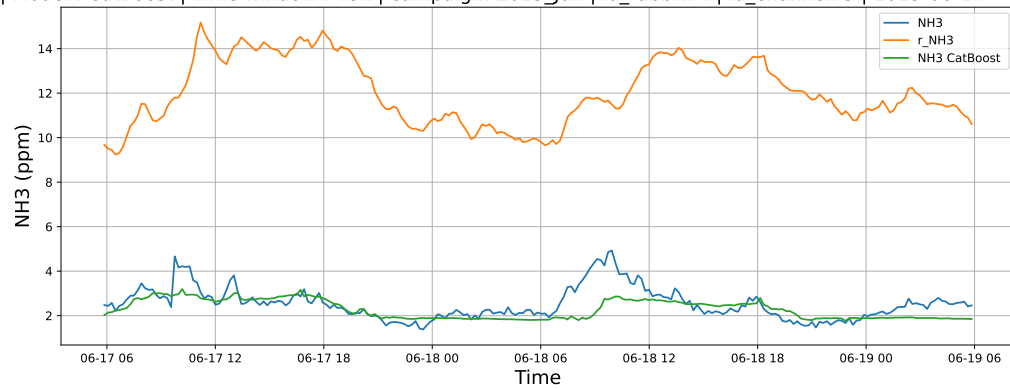
X does not have valid feature names, but LGBMRegressor was fitted with feature names

NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



3.1.3.2 Time window : 48

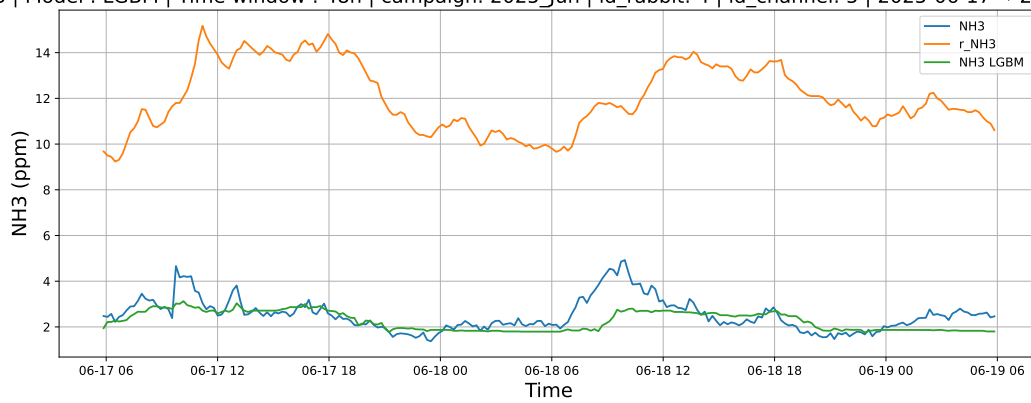
NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



C:\Users\titou\anaconda3\Lib\site-packages\sklearn\utils\validation.py:2749: UserWarning:

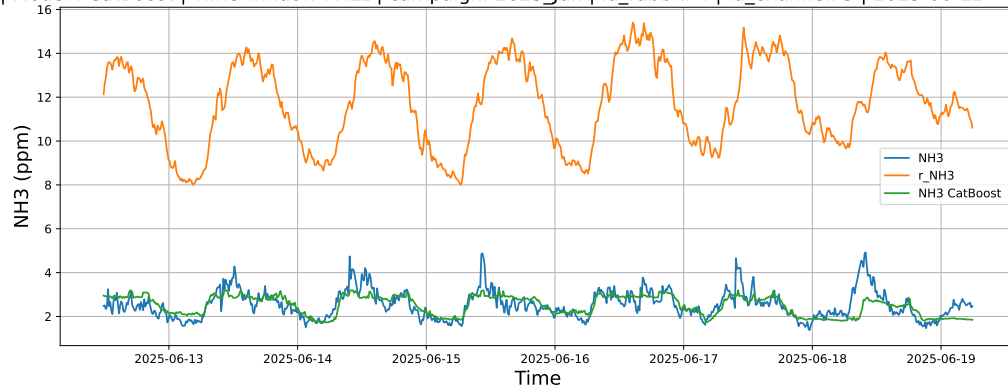
X does not have valid feature names, but LGBMRegressor was fitted with feature names

NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



3.1.3.3 Time window : ALL

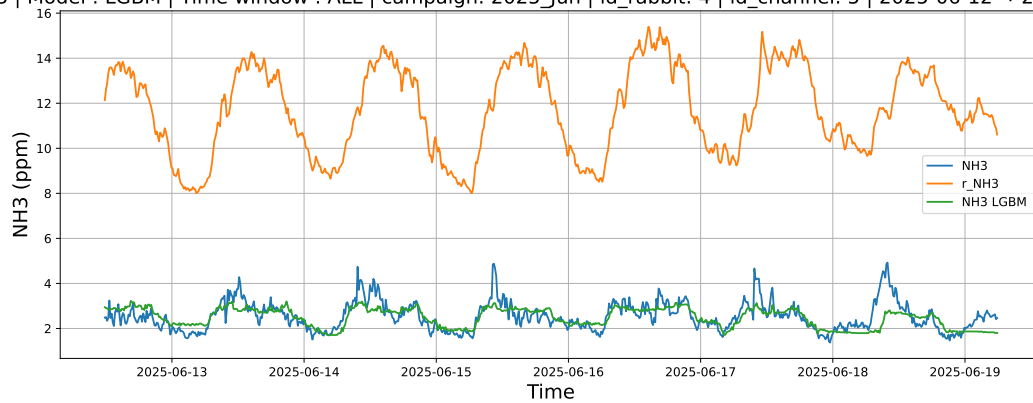
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



C:\Users\titou\anaconda3\Lib\site-packages\sklearn\utils\validation.py:2749: UserWarning:

X does not have valid feature names, but LGBMRegressor was fitted with feature names

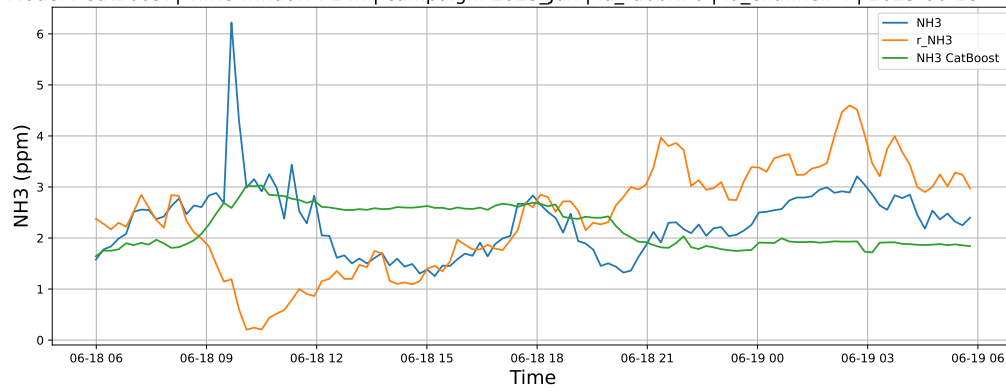
NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



3.1.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

3.1.4.1 Time window : 24

NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



C:\Users\titou\anaconda3\Lib\site-packages\sklearn\utils\validation.py:2749: UserWarning:

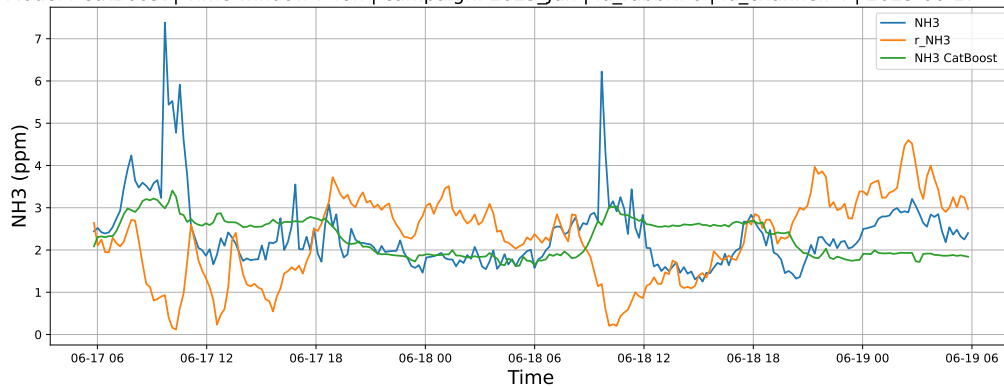
X does not have valid feature names, but LGBMRegressor was fitted with feature names

NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



3.1.4.2 Time window : 48

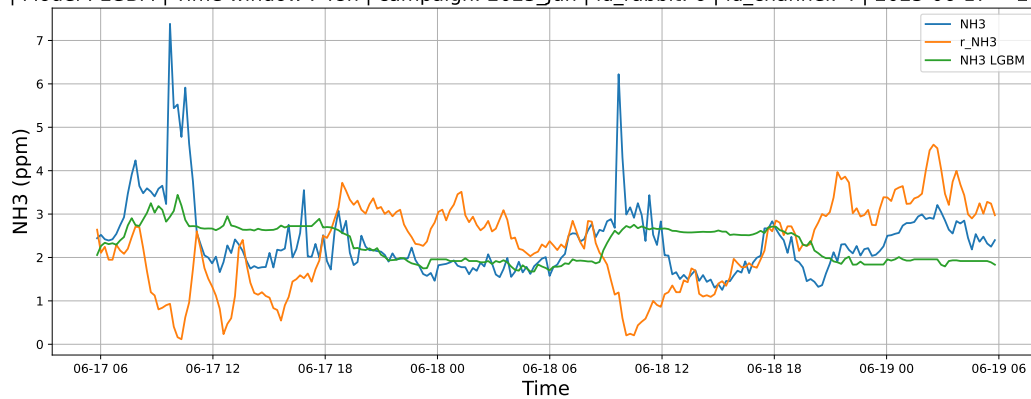
NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



C:\Users\titou\anaconda3\Lib\site-packages\sklearn\utils\validation.py:2749: UserWarning:

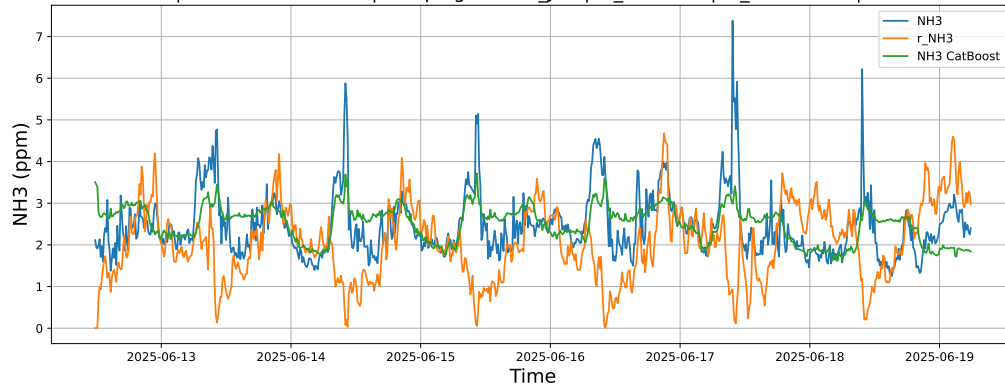
X does not have valid feature names, but LGBMRegressor was fitted with feature names

NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



3.1.4.3 Time window : ALL

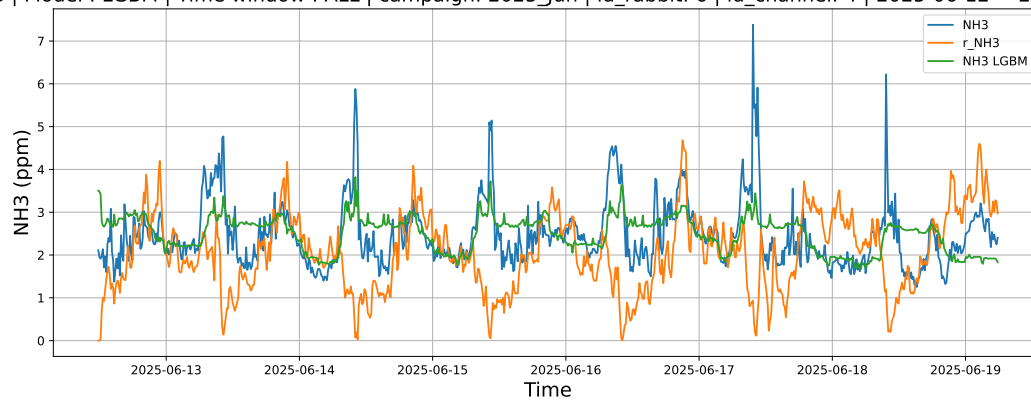
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



C:\Users\titou\anaconda3\Lib\site-packages\sklearn\utils\validation.py:2749: UserWarning:

X does not have valid feature names, but LGBMRegressor was fitted with feature names

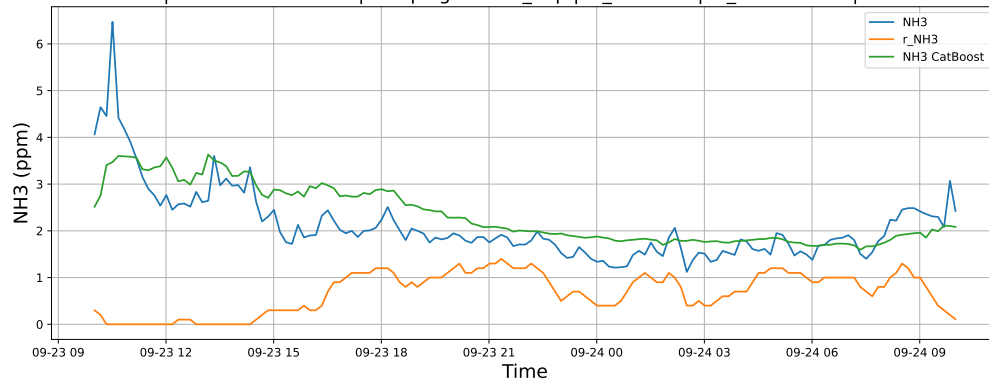
NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



3.1.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

3.1.5.1 Time window : 24

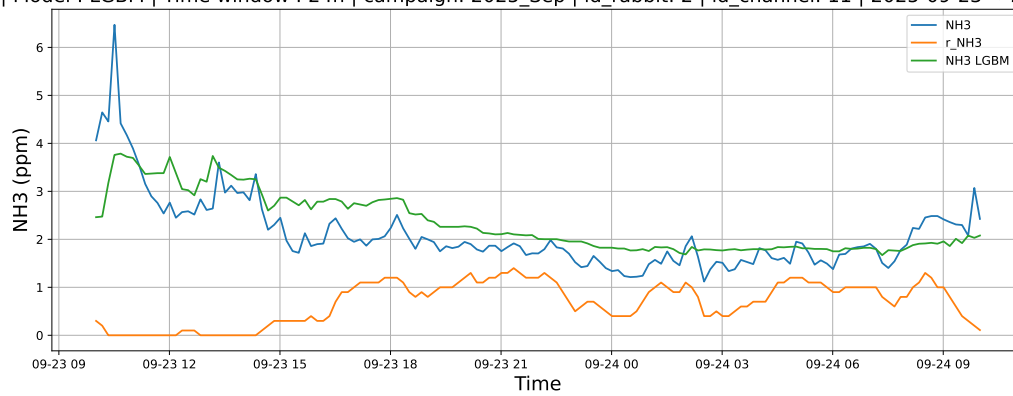
NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



C:\Users\titou\anaconda3\Lib\site-packages\sklearn\utils\validation.py:2749: UserWarning:

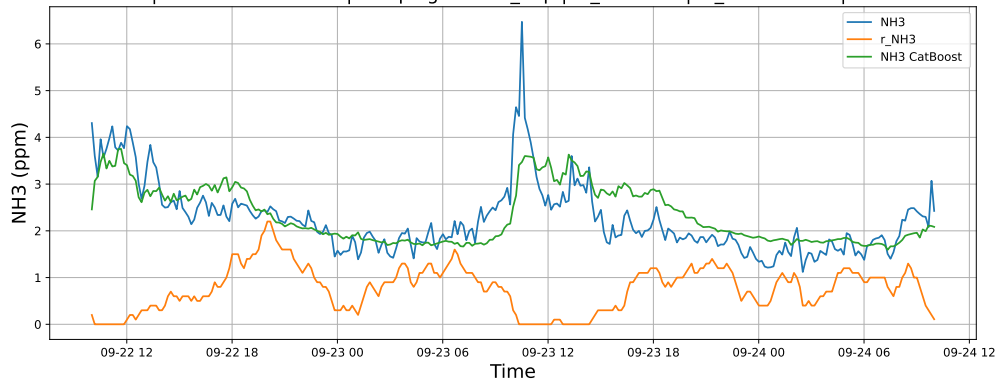
X does not have valid feature names, but LGBMRegressor was fitted with feature names

NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



3.1.5.2 Time window : 48

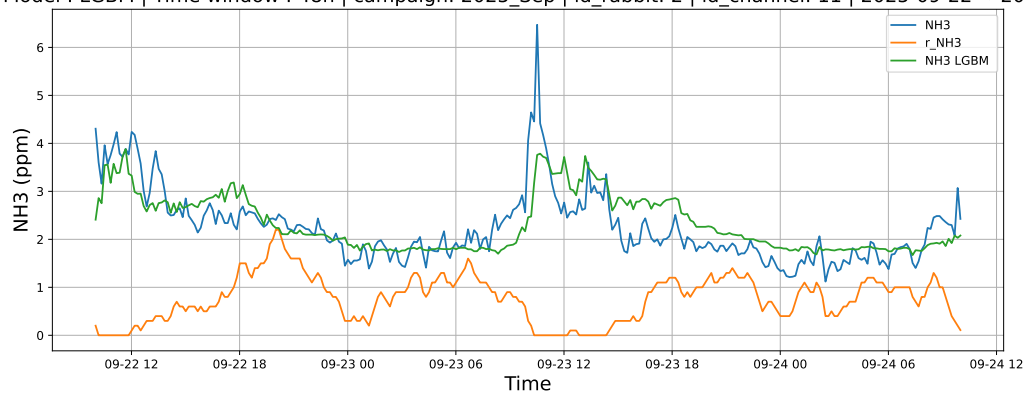
NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



C:\Users\titou\anaconda3\Lib\site-packages\sklearn\utils\validation.py:2749: UserWarning:

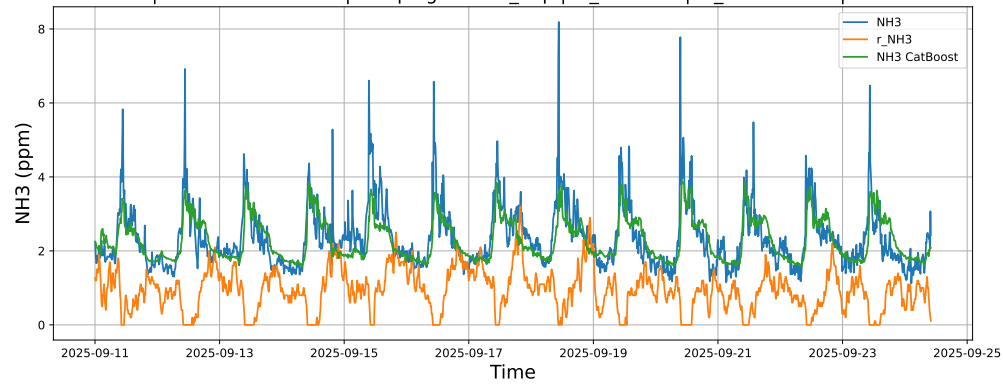
X does not have valid feature names, but LGBMRegressor was fitted with feature names

NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



3.1.5.3 Time window : ALL

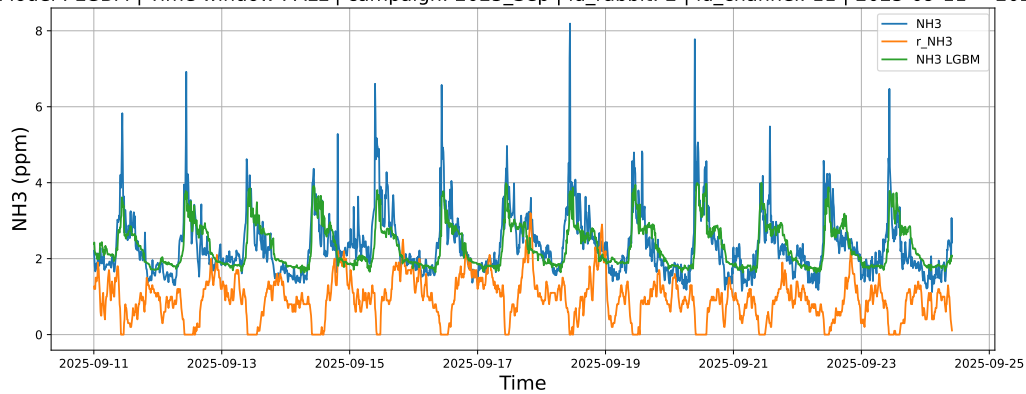
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



C:\Users\titou\anaconda3\Lib\site-packages\sklearn\utils\validation.py:2749: UserWarning:

X does not have valid feature names, but LGBMRegressor was fitted with feature names

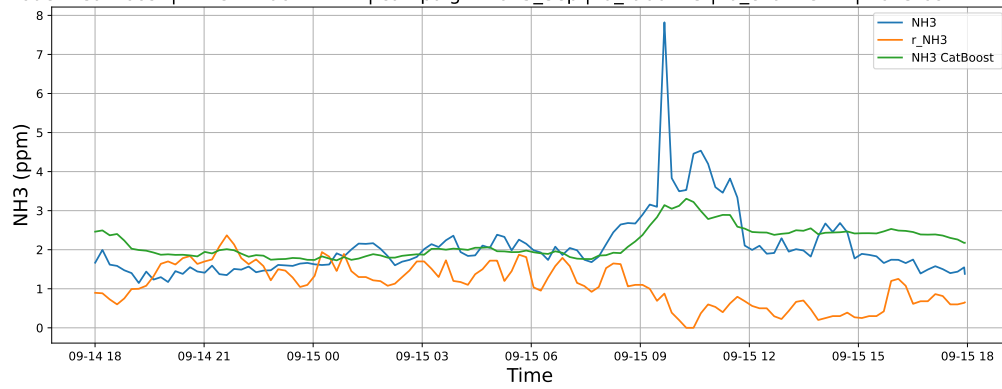
NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



3.1.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

3.1.6.1 Time window : 24

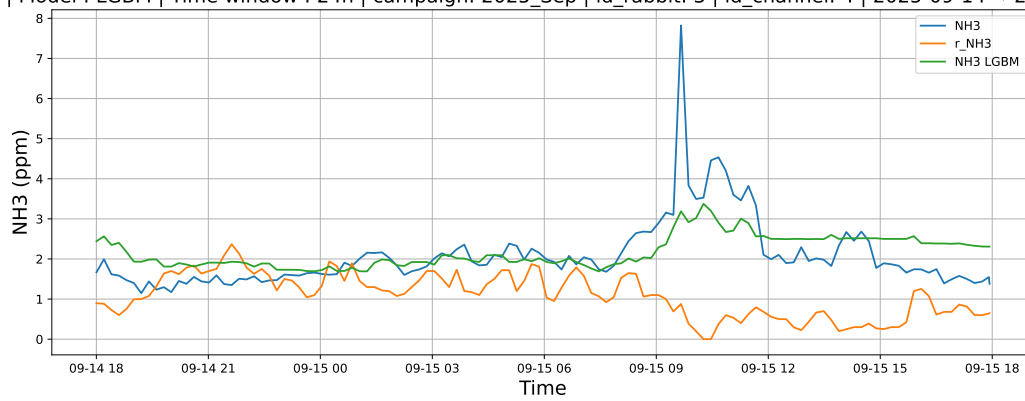
NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



C:\Users\titou\anaconda3\Lib\site-packages\sklearn\utils\validation.py:2749: UserWarning:

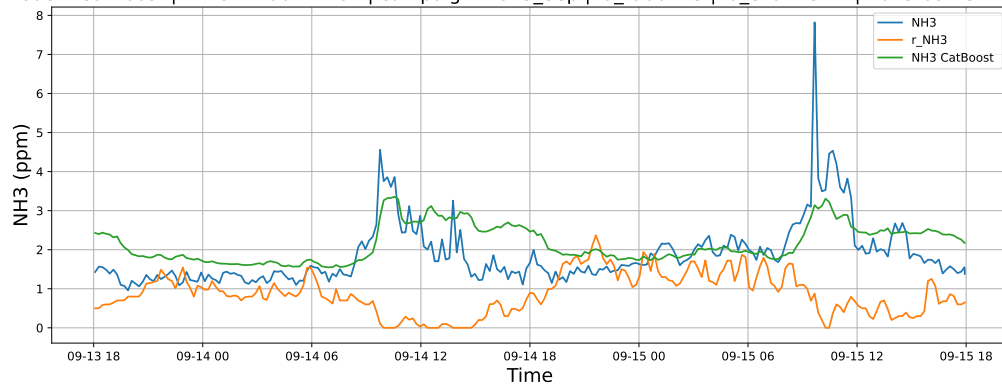
X does not have valid feature names, but LGBMRegressor was fitted with feature names

NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



3.1.6.2 Time window : 48

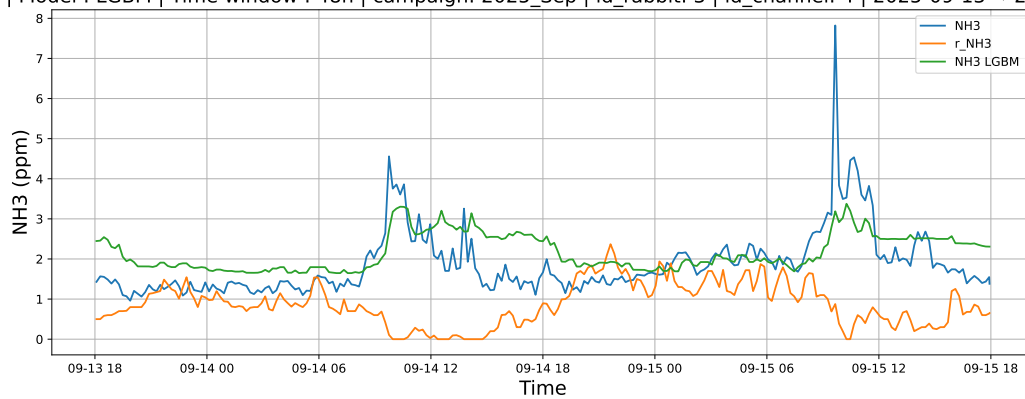
NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



C:\Users\titou\anaconda3\Lib\site-packages\sklearn\utils\validation.py:2749: UserWarning:

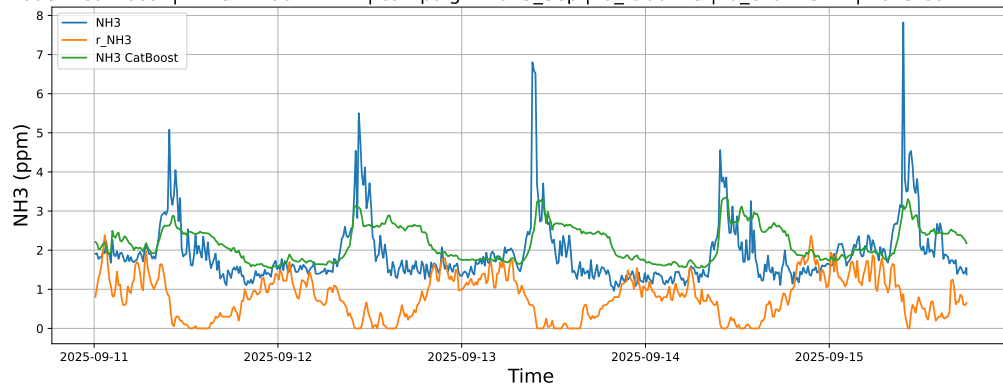
X does not have valid feature names, but LGBMRegressor was fitted with feature names

NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



3.1.6.3 Time window : ALL

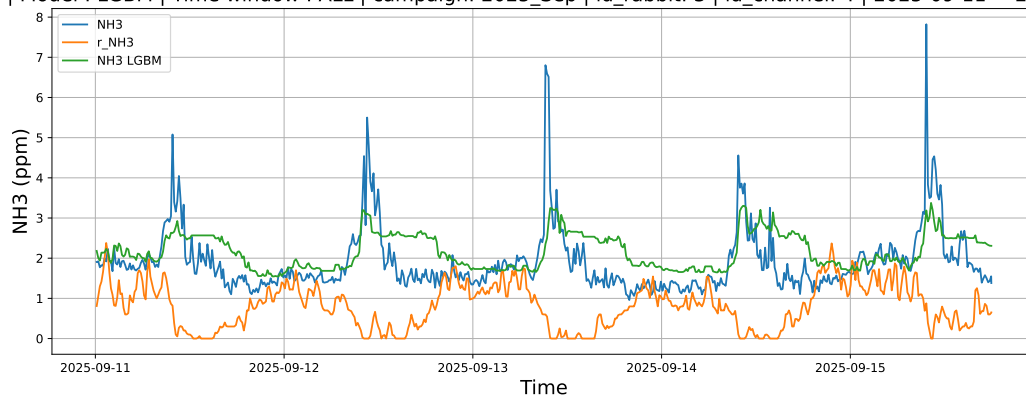
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



C:\Users\titou\anaconda3\Lib\site-packages\sklearn\utils\validation.py:2749: UserWarning:

X does not have valid feature names, but LGBMRegressor was fitted with feature names

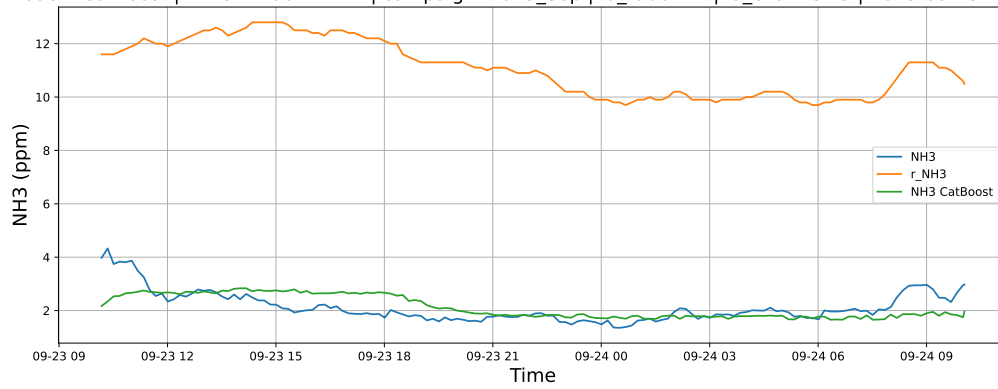
NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



3.1.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

3.1.7.1 Time window : 24

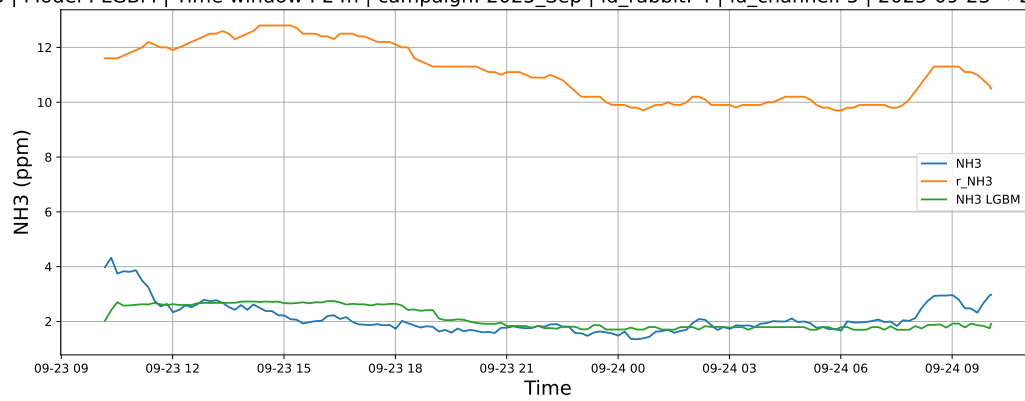
NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



C:\Users\titou\anaconda3\Lib\site-packages\sklearn\utils\validation.py:2749: UserWarning:

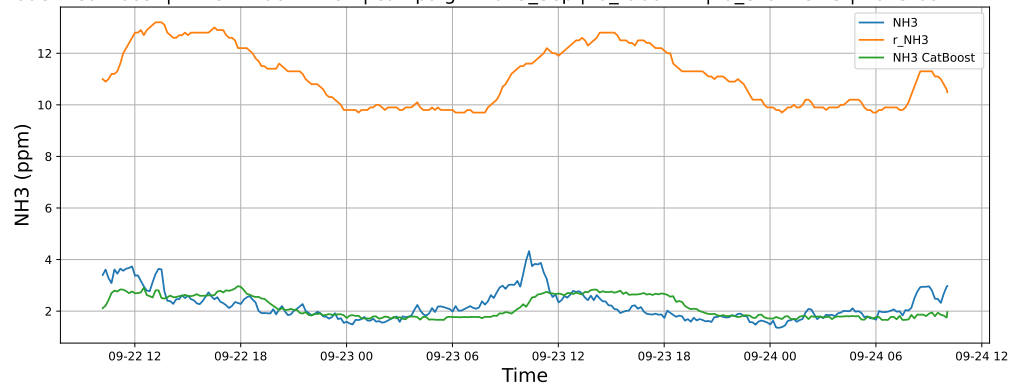
X does not have valid feature names, but LGBMRegressor was fitted with feature names

NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



3.1.7.2 Time window : 48

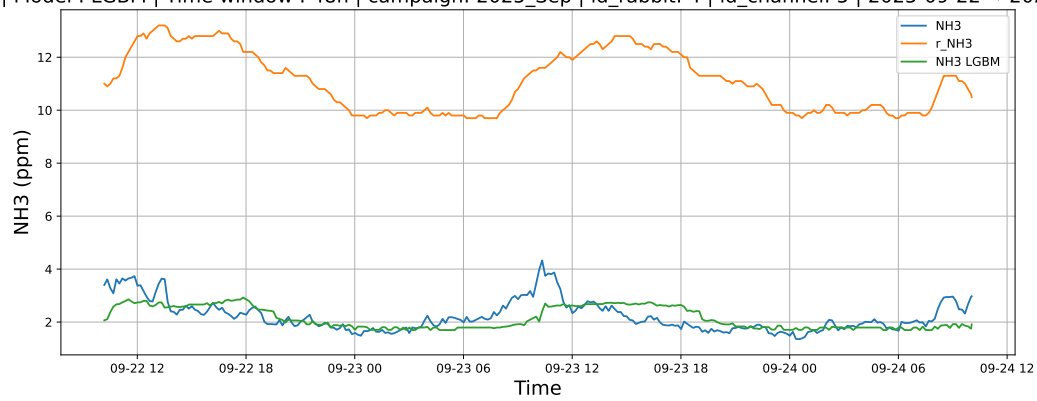
NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



C:\Users\titou\anaconda3\Lib\site-packages\sklearn\utils\validation.py:2749: UserWarning:

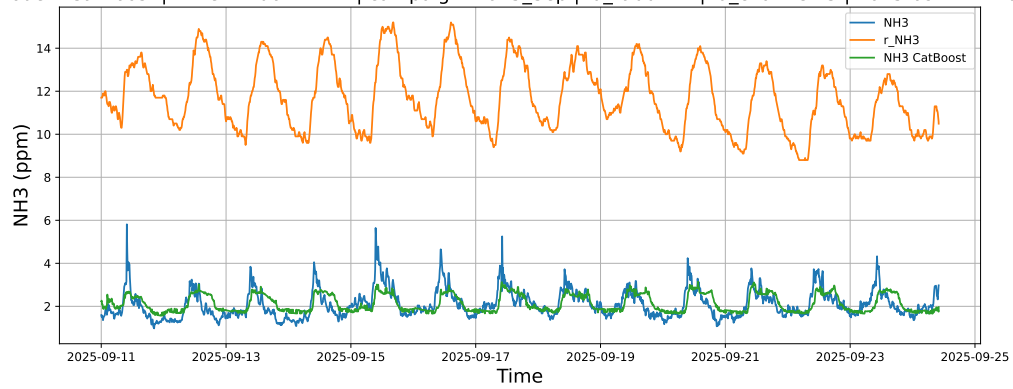
X does not have valid feature names, but LGBMRegressor was fitted with feature names

NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



3.1.7.3 Time window : ALL

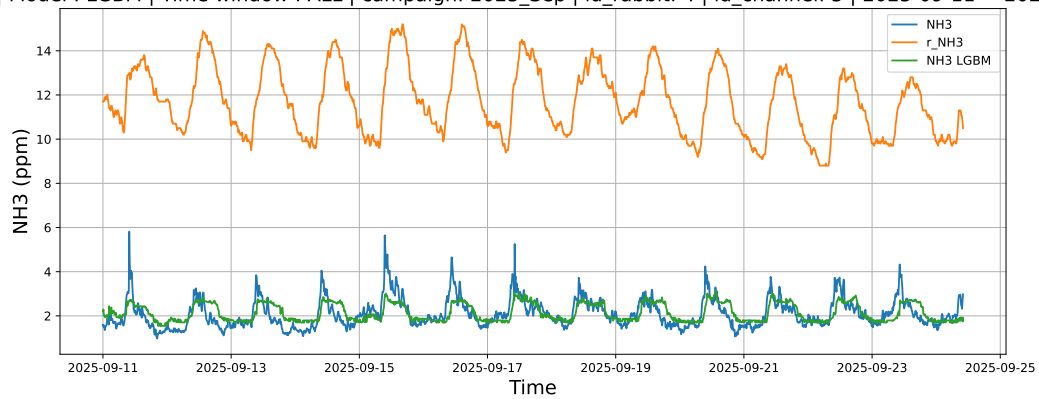
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



C:\Users\titou\anaconda3\Lib\site-packages\sklearn\utils\validation.py:2749: UserWarning:

X does not have valid feature names, but LGBMRegressor was fitted with feature names

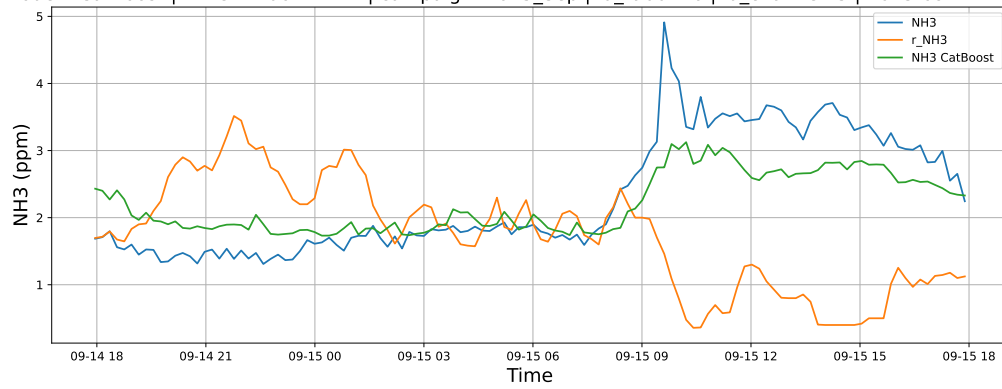
NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



3.1.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

3.1.8.1 Time window : 24

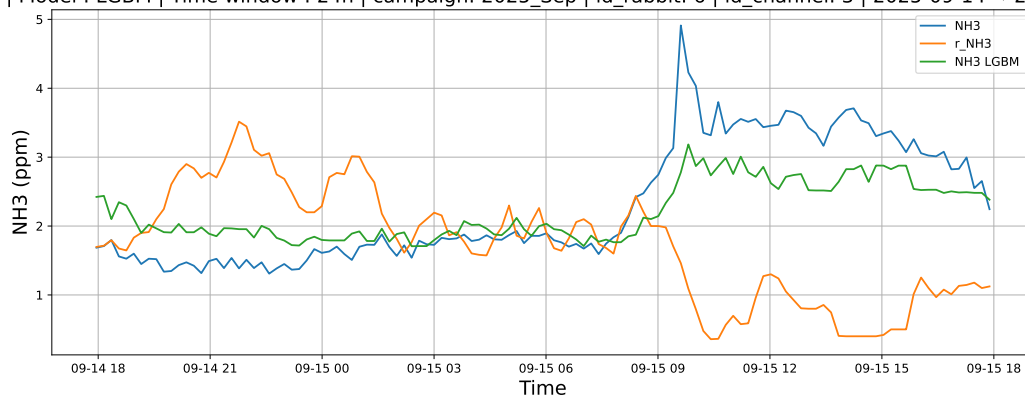
NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



C:\Users\titou\anaconda3\Lib\site-packages\sklearn\utils\validation.py:2749: UserWarning:

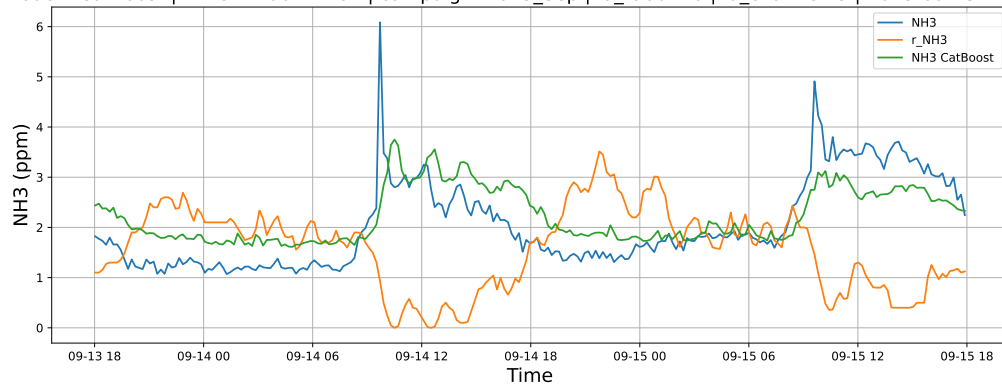
X does not have valid feature names, but LGBMRegressor was fitted with feature names

NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



3.1.8.2 Time window : 48

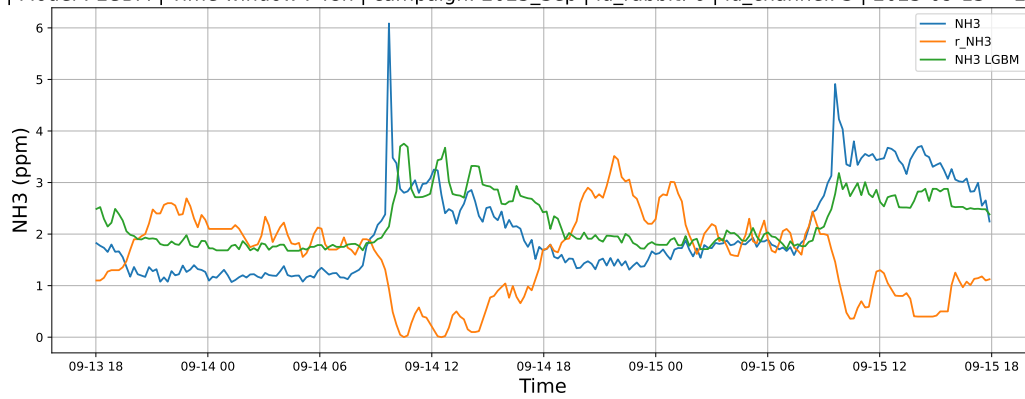
NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



C:\Users\titou\anaconda3\Lib\site-packages\sklearn\utils\validation.py:2749: UserWarning:

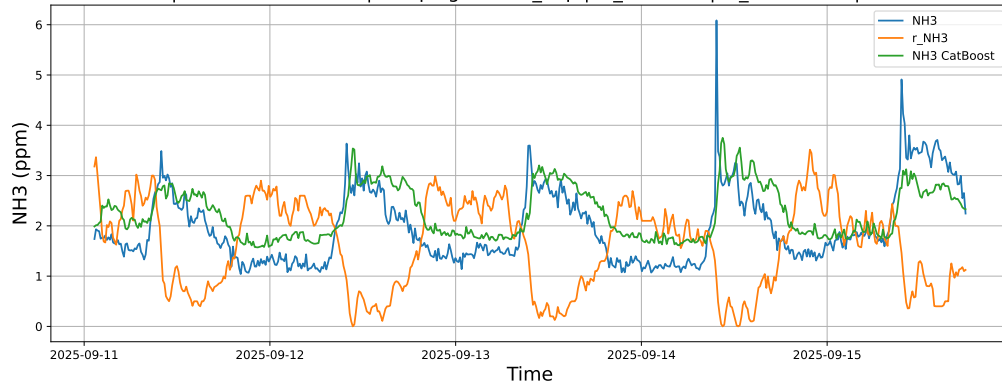
X does not have valid feature names, but LGBMRegressor was fitted with feature names

NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



3.1.8.3 Time window : ALL

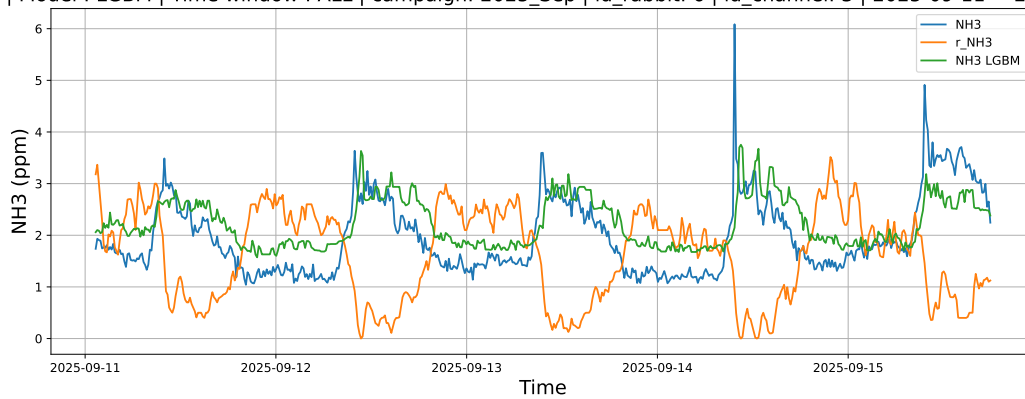
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



C:\Users\titou\anaconda3\Lib\site-packages\sklearn\utils\validation.py:2749: UserWarning:

X does not have valid feature names, but LGBMRegressor was fitted with feature names

NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



3.2 Compare plots

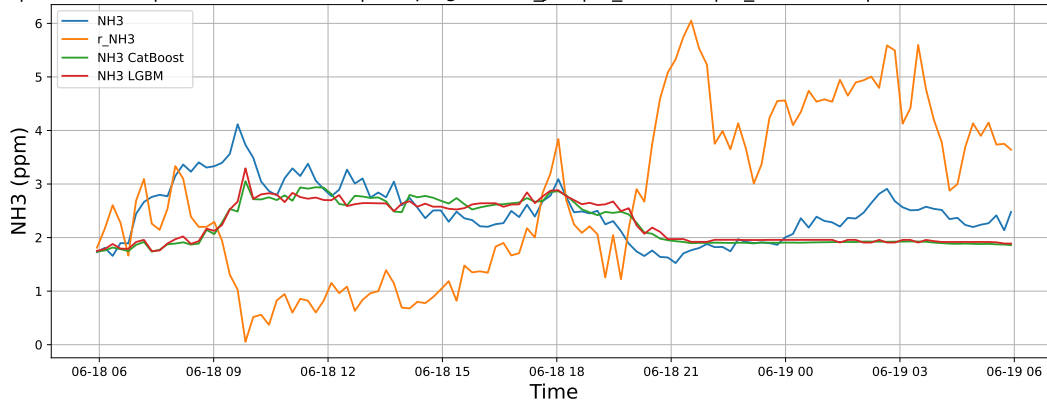
3.2.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

3.2.1.1 Time window : 24

C:\Users\titou\anaconda3\Lib\site-packages\sklearn\utils\validation.py:2749: UserWarning:

X does not have valid feature names, but LGBMRegressor was fitted with feature names

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19

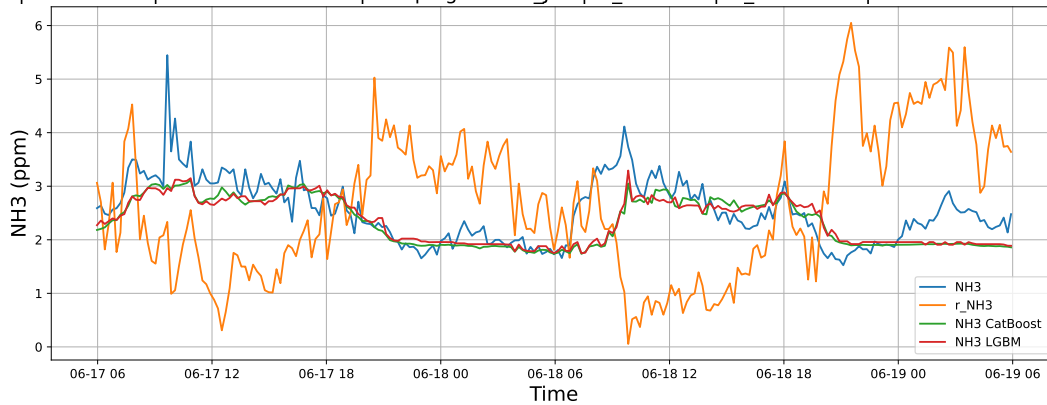


3.2.1.2 Time window : 48

C:\Users\titou\anaconda3\Lib\site-packages\sklearn\utils\validation.py:2749: UserWarning:

X does not have valid feature names, but LGBMRegressor was fitted with feature names

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

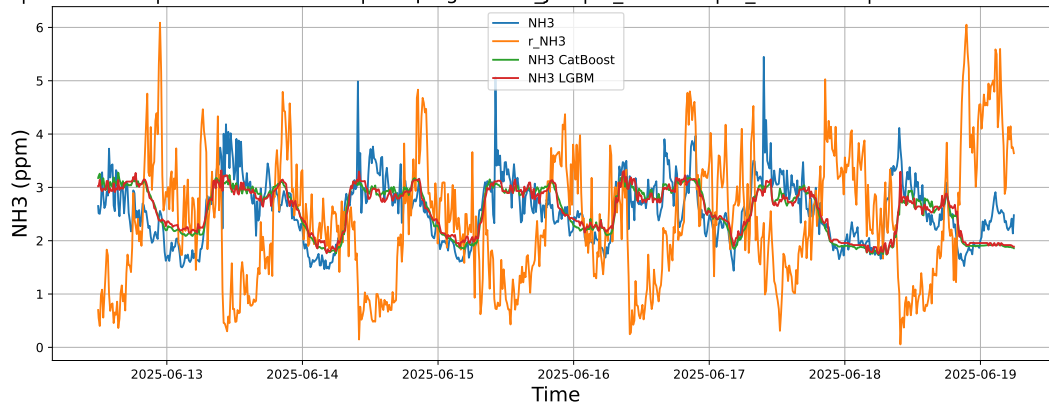


3.2.1.3 Time window : ALL

C:\Users\titou\anaconda3\Lib\site-packages\sklearn\utils\validation.py:2749: UserWarning:

X does not have valid feature names, but LGBMRegressor was fitted with feature names

NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



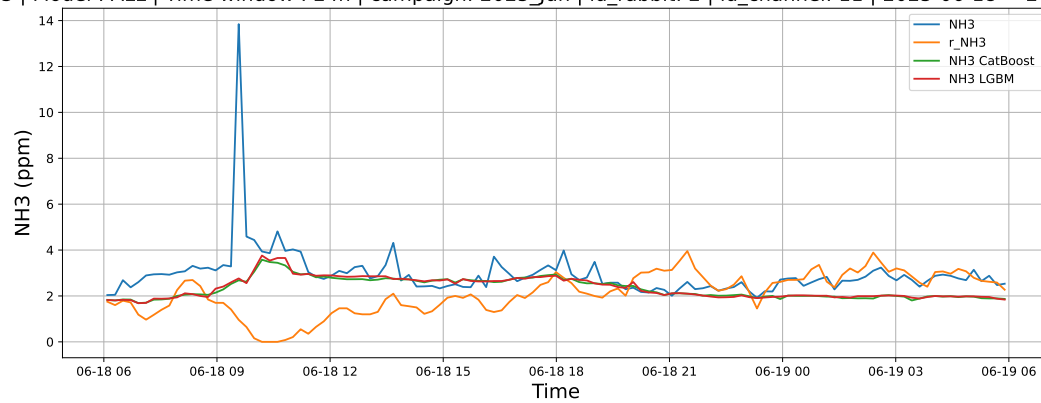
3.2.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

3.2.2.1 Time window : 24

C:\Users\titou\anaconda3\Lib\site-packages\sklearn\utils\validation.py:2749: UserWarning:

X does not have valid feature names, but LGBMRegressor was fitted with feature names

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

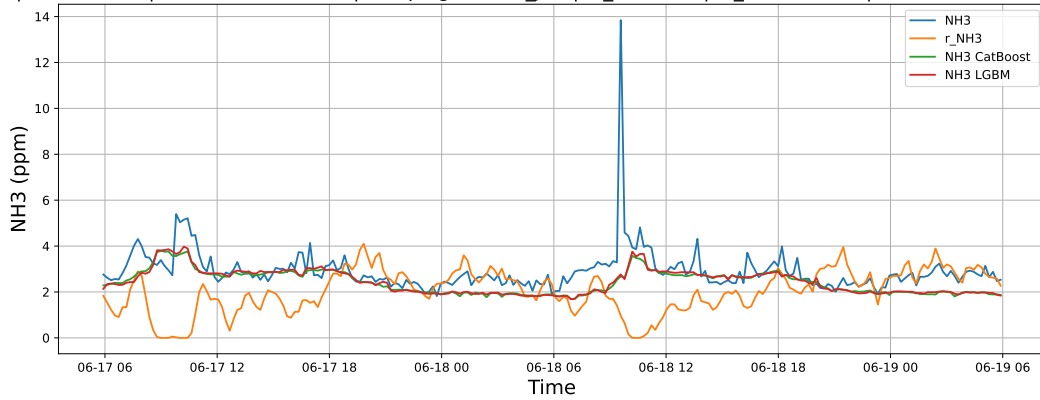


3.2.2.2 Time window : 48

C:\Users\titou\anaconda3\Lib\site-packages\sklearn\utils\validation.py:2749: UserWarning:

X does not have valid feature names, but LGBMRegressor was fitted with feature names

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

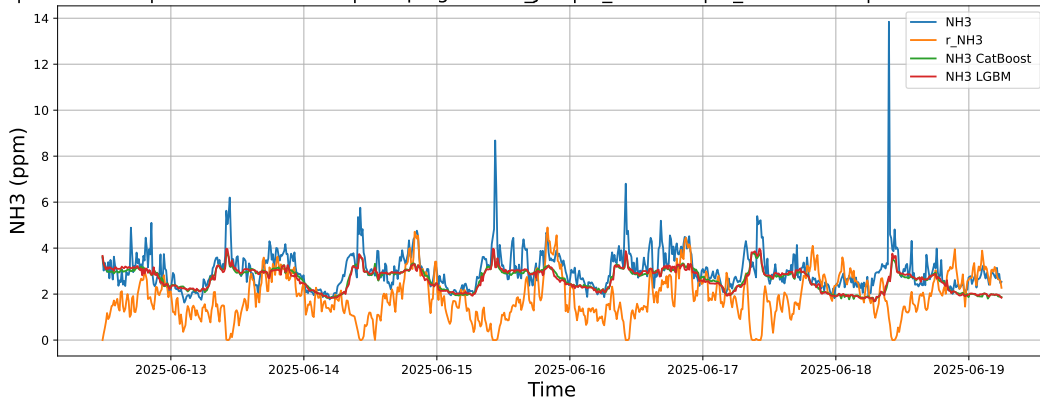


3.2.2.3 Time window : ALL

C:\Users\titou\anaconda3\Lib\site-packages\sklearn\utils\validation.py:2749: UserWarning:

X does not have valid feature names, but LGBMRegressor was fitted with feature names

NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



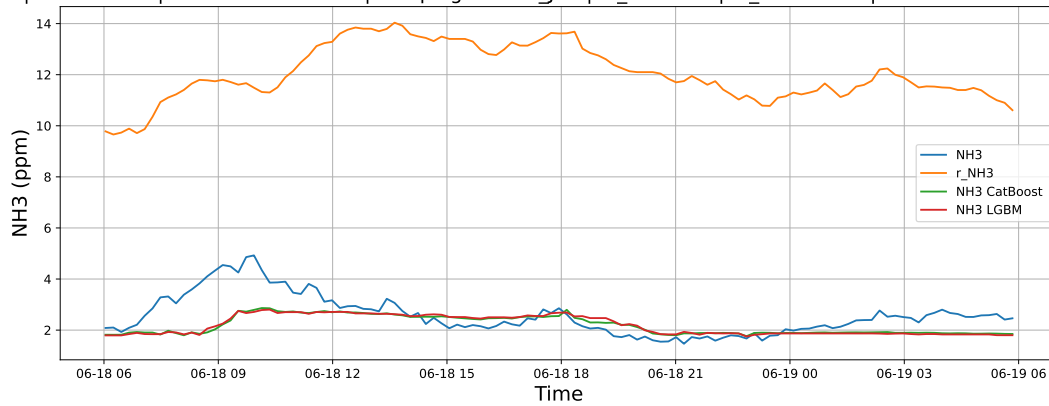
3.2.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

3.2.3.1 Time window : 24

C:\Users\titou\anaconda3\Lib\site-packages\sklearn\utils\validation.py:2749: UserWarning:

X does not have valid feature names, but LGBMRegressor was fitted with feature names

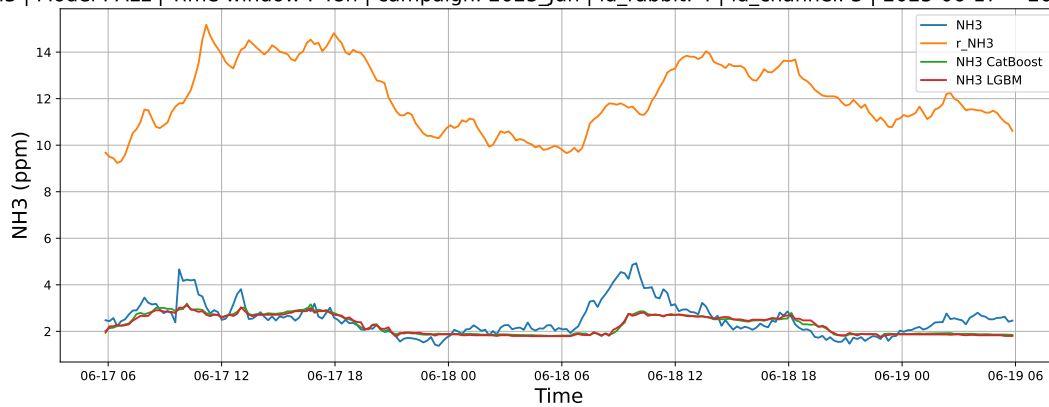
NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



3.2.3.2 Time window : 48

C:\Users\titou\anaconda3\Lib\site-packages\sklearn\utils\validation.py:2749: UserWarning:
X does not have valid feature names, but LGBMRegressor was fitted with feature names

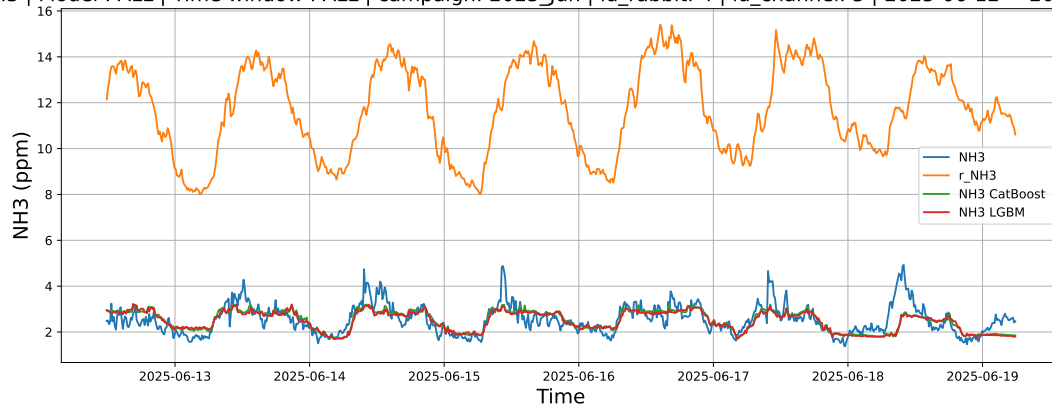
NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



3.2.3.3 Time window : ALL

C:\Users\titou\anaconda3\Lib\site-packages\sklearn\utils\validation.py:2749: UserWarning:
X does not have valid feature names, but LGBMRegressor was fitted with feature names

NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



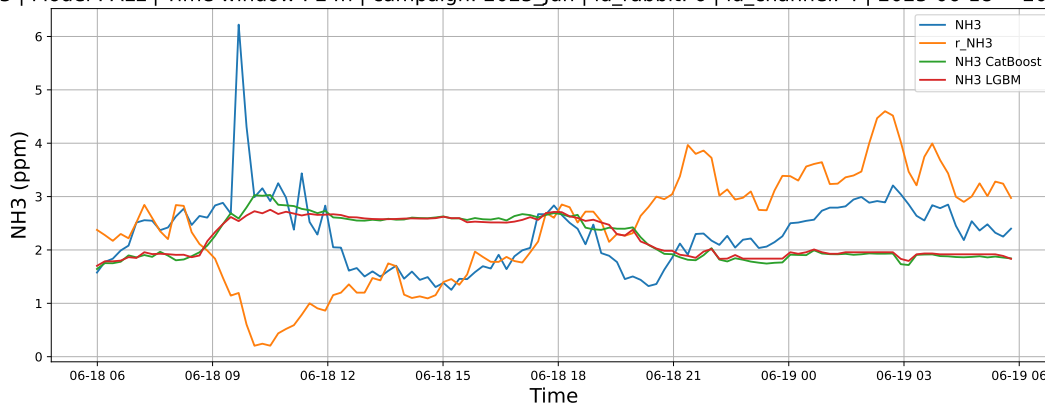
3.2.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

3.2.4.1 Time window : 24

C:\Users\titou\anaconda3\Lib\site-packages\sklearn\utils\validation.py:2749: UserWarning:

X does not have valid feature names, but LGBMRegressor was fitted with feature names

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

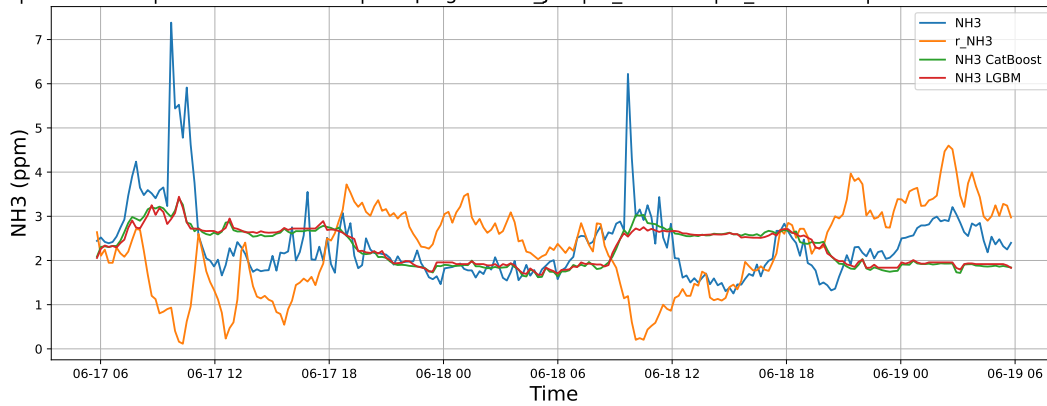


3.2.4.2 Time window : 48

C:\Users\titou\anaconda3\Lib\site-packages\sklearn\utils\validation.py:2749: UserWarning:

X does not have valid feature names, but LGBMRegressor was fitted with feature names

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

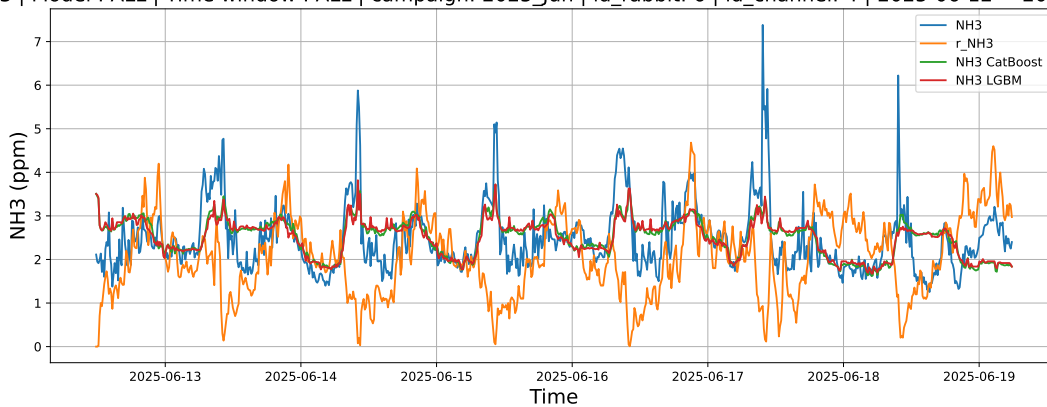


3.2.4.3 Time window : ALL

C:\Users\titou\anaconda3\Lib\site-packages\sklearn\utils\validation.py:2749: UserWarning:

X does not have valid feature names, but LGBMRegressor was fitted with feature names

NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



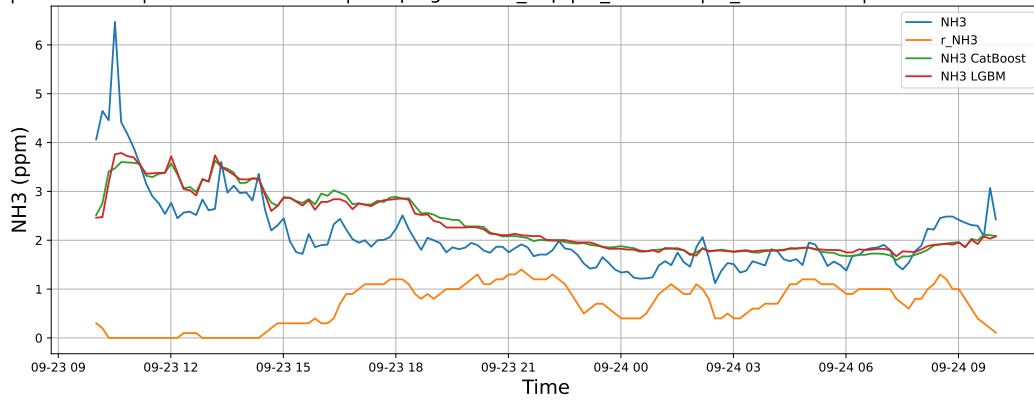
3.2.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

3.2.5.1 Time window : 24

C:\Users\titou\anaconda3\Lib\site-packages\sklearn\utils\validation.py:2749: UserWarning:

X does not have valid feature names, but LGBMRegressor was fitted with feature names

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

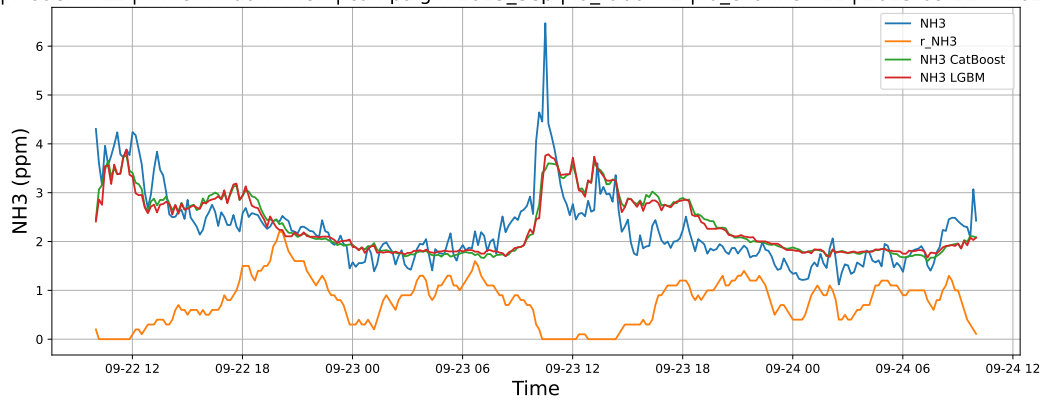


3.2.5.2 Time window : 48

C:\Users\titou\anaconda3\Lib\site-packages\sklearn\utils\validation.py:2749: UserWarning:

X does not have valid feature names, but LGBMRegressor was fitted with feature names

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

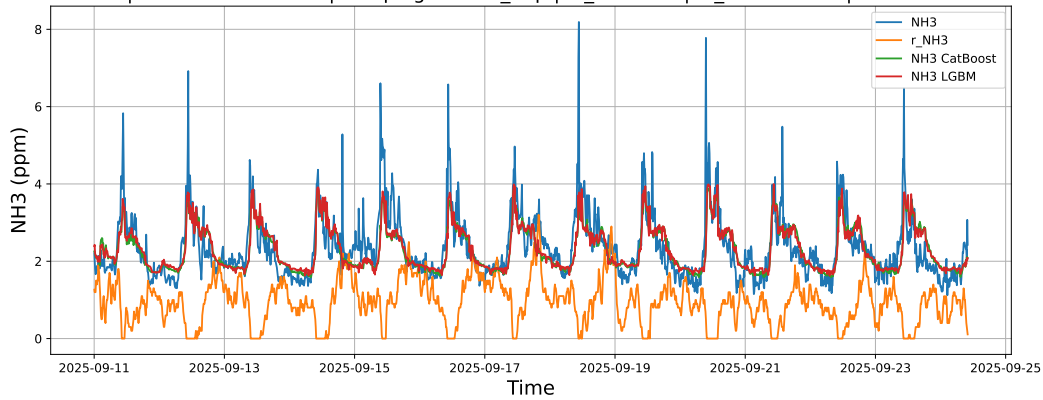


3.2.5.3 Time window : ALL

C:\Users\titou\anaconda3\Lib\site-packages\sklearn\utils\validation.py:2749: UserWarning:

X does not have valid feature names, but LGBMRegressor was fitted with feature names

NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



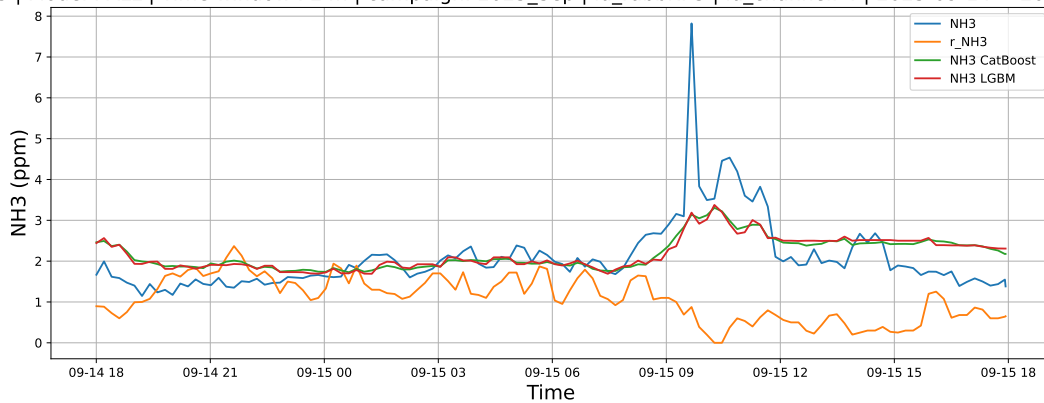
3.2.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

3.2.6.1 Time window : 24

C:\Users\titou\anaconda3\Lib\site-packages\sklearn\utils\validation.py:2749: UserWarning:

X does not have valid feature names, but LGBMRegressor was fitted with feature names

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

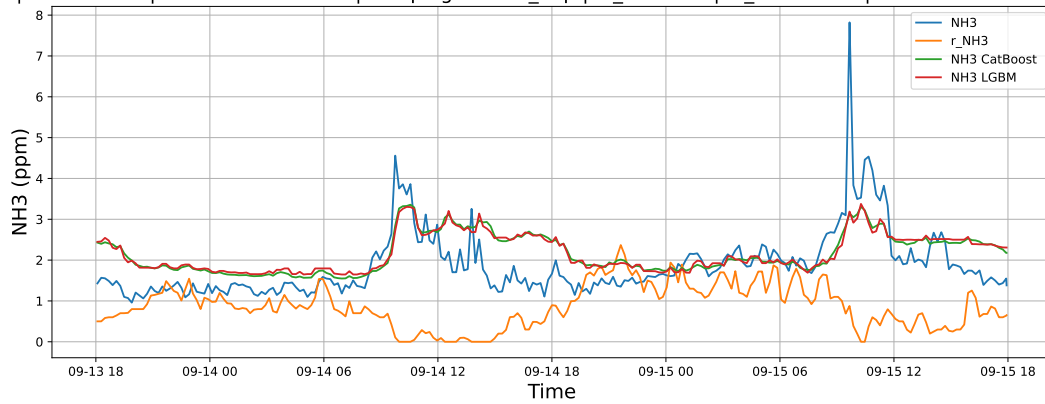


3.2.6.2 Time window : 48

C:\Users\titou\anaconda3\Lib\site-packages\sklearn\utils\validation.py:2749: UserWarning:

X does not have valid feature names, but LGBMRegressor was fitted with feature names

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

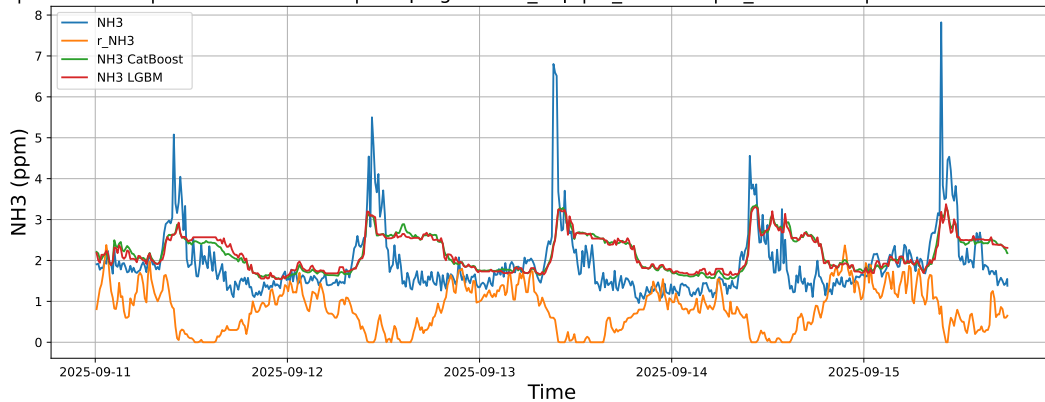


3.2.6.3 Time window : ALL

C:\Users\titou\anaconda3\Lib\site-packages\sklearn\utils\validation.py:2749: UserWarning:

X does not have valid feature names, but LGBMRegressor was fitted with feature names

NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



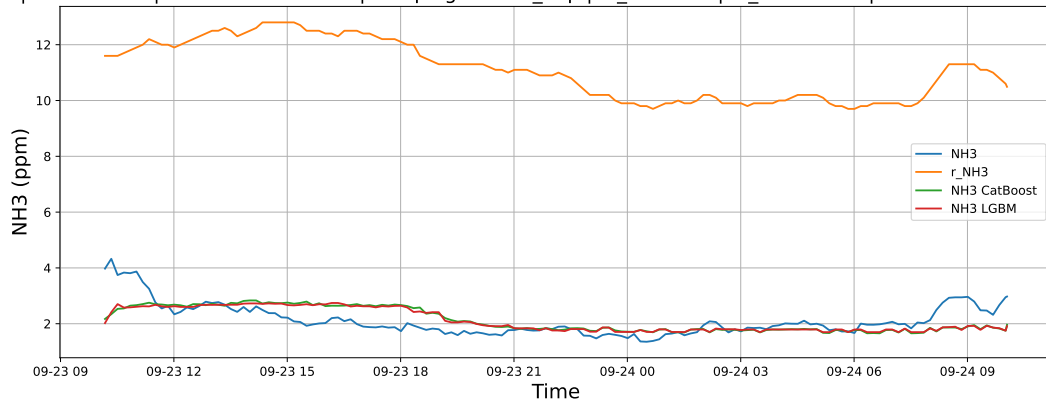
3.2.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

3.2.7.1 Time window : 24

C:\Users\titou\anaconda3\Lib\site-packages\sklearn\utils\validation.py:2749: UserWarning:

X does not have valid feature names, but LGBMRegressor was fitted with feature names

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24

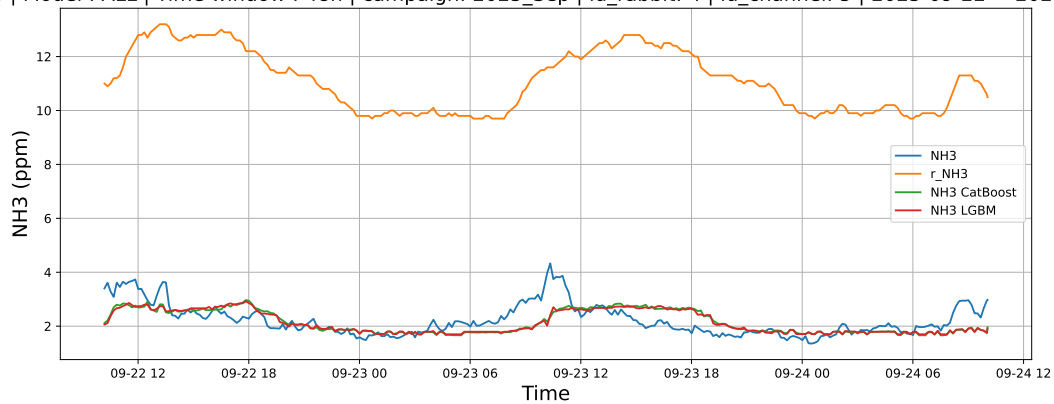


3.2.7.2 Time window : 48

C:\Users\titou\anaconda3\Lib\site-packages\sklearn\utils\validation.py:2749: UserWarning:

X does not have valid feature names, but LGBMRegressor was fitted with feature names

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

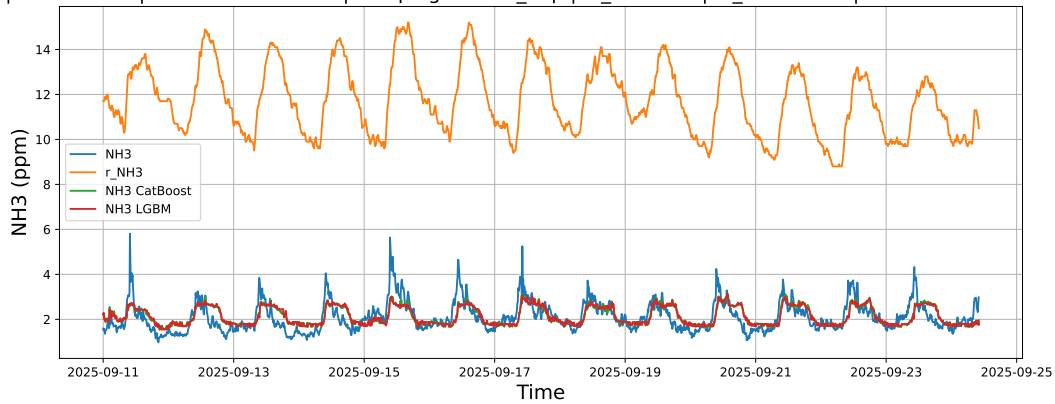


3.2.7.3 Time window : ALL

C:\Users\titou\anaconda3\Lib\site-packages\sklearn\utils\validation.py:2749: UserWarning:

X does not have valid feature names, but LGBMRegressor was fitted with feature names

NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



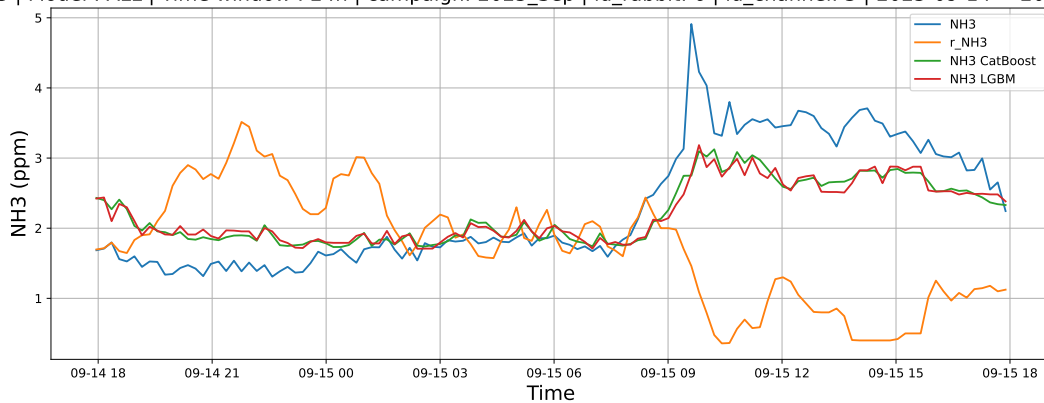
3.2.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

3.2.8.1 Time window : 24

C:\Users\titou\anaconda3\Lib\site-packages\sklearn\utils\validation.py:2749: UserWarning:

X does not have valid feature names, but LGBMRegressor was fitted with feature names

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

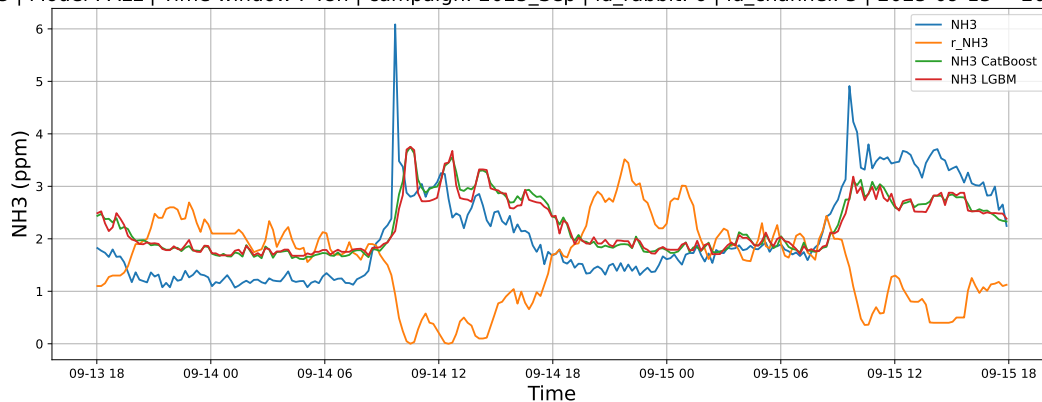


3.2.8.2 Time window : 48

C:\Users\titou\anaconda3\Lib\site-packages\sklearn\utils\validation.py:2749: UserWarning:

X does not have valid feature names, but LGBMRegressor was fitted with feature names

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



3.2.8.3 Time window : ALL

C:\Users\titou\anaconda3\Lib\site-packages\sklearn\utils\validation.py:2749: UserWarning:

X does not have valid feature names, but LGBMRegressor was fitted with feature names

NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15

